# Finitely Generated Matrix Convex Sets and Classification of Operator Systems Generated by $\Lambda$-Commuting Unitaries

**Ran Kiri**

# Finitely Generated Matrix Convex Sets and Classification of Operator Systems Generated by $\Lambda$-Commuting Unitaries

Research Thesis

Submitted in partial fulfillment of the requirements
for the degree of Master of Science in Mathematics

**Ran Kiri**

Submitted to the Senate
of the Technion — Israel Institute of Technology Av 5782
Haifa August 2022

This research was carried out under the supervision of Prof. Orr Shalit, in the Faculty of Mathematics.

The generous financial help of the Technion is gratefully acknowledged.

# Contents


**1 Introduction** **2**

**2 Preliminaries and notation** **4**
2.1 Operator systems and positivity . . . . . 4
2.2 *-homomorphisms, positive maps and completely positive maps . . . . 4
2.3 Matrix convex sets . . . . . 7
2.3.1 Matrix extreme points . . . . . 10
2.3.2 Matrix affine functions . . . . . 12

**3 $N$-generated compact matrix convex sets** **14**
3.1 Characterization of matrix extreme points . . . . . 14
3.1.1 $N$-minimal matrix convex sets . . . . . 17
3.2 $N$-generated matrix state spaces . . . . . 18
3.3 $N$-generated matrix ranges . . . . . 21

**4 Operator Systems Generated by Unitaries** **25**
4.1 Order Equivalence and $*$-Isomorphisms . . . . . 25
4.2 Tuple Equivalence Using Dilation Theory . . . . . 26

**5 Rotation Algebras and $q$-Commuting Unitaries** **28**
5.1 Rational Rotation Algebras . . . . . 30
5.2 Irrational rotation algebras . . . . . 34
5.2.1 Equivalence in the class of $q$-commutation . . . . . 35
5.3 Rational $\Lambda$-commuting $d$-tuples . . . . . 36
5.4 The isometric case . . . . . 38

**6 Concluding remarks** **41**
6.1 Compact matrix convex sets . . . . . 41
6.2 Representations of $\Lambda$-commuting $d$-tuples . . . . . 41

# List of Symbols

| | |
|---|---|
| $\boldsymbol{\theta}, \boldsymbol{\phi}$ | Matrix affine functions between two matrix convex sets. |
| $\mathbf{s}, \mathbf{r}$ | $d$-tuple of operators. |
| $\mathcal{A}, \mathcal{B}$ | Unital $C^*$-algebra. |
| $\mathcal{H}, \mathcal{K}$ | Complex Hilbert space. |
| $\mathcal{S}(\mathbf{s})$ | The operator system generated by a $d$-tuple $\mathbf{s}$. |
| $\mathcal{S}, \mathcal{R}$ | Operator system. |
| $\Lambda$ | $d \times d$ self-adjoint matrix with complex entries of modulus 1. |
| $\mathbb{B}_d$ | The euclidean unit ball in $\mathbb{R}^d$. |
| $\mathbb{C}$ | The complex numbers. |
| $\mathbb{N}$ | The natural numbers. |
| $\mathbb{R}$ | The real numbers. |
| $\mathcal{L}, \mathcal{C}$ | Matrix convex sets of complex matrices. |
| $\mathbb{1}_n$ | The unit $n \times n$ matrix. |
| $\mathbb{1}_{\mathcal{S}}, \mathbb{1}_{\mathcal{R}}$ | Unit element in an operator system. |
| $\mathbb{M}_{n,m}, \mathbb{M}_n$ | The set of $n \times m$ or $n \times n$ matrices with entries from $\mathbb{C}$. |
| $\mathbb{M}_{n,m}(V), \mathbb{M}_n(V)$ | The set of $n \times m$ or $n \times n$ matrices with entries from a some vector space $V$. |
| $\partial K, \partial K_n$ | The set of matrix extreme points of $K$ or the set of matrix extreme points which lie in $K_n$. |
| $\pi, \rho$ | $*$-homomorphism. |
| $\mathrm{UCP}(\mathcal{S})$ | Matrix state space of the operator system $\mathcal{S}$. |
| $\varphi, \phi, \psi, \Phi$ | Linear map. |
| $\mathcal{W}(\mathbf{s})$ | Matrix range of the $d$-tuple $\mathbf{s}$. |

| | |
|---|---|
| $A(K)$ | Set of matrix affine functions from $K$ to $(\mathbb{M}_n)_{n=1}^{\infty}$. |
| $B(\mathcal{H}), B(\mathcal{K}, \mathcal{H})$ | Bounded operators on a Hilbert space $\mathcal{H}$ or from a Hilbert space $\mathcal{H}$ to a Hilbert space $\mathcal{K}$. |
| $C^*(\mathbf{s}), C^*(\mathcal{S})$ | The $C^*$-algebra generated by a $d$-tuple $\mathbf{s}$ or by an operator system $\mathcal{S}$. |
| $F$ | Matrix affine function from matrix convex set to $(\mathbb{M}_n)_{n=1}^{\infty}$. |
| $K$ | Compact matrix convex set over some locally convex vector space $V$. |
| $L(V, W)$ | The set of linear maps from $V$ to $W$. |
| $q$ | Complex number of modulus 1. |
| $s, r, s_i, r_i$ | Element in an operator system. |
| $V$ | Vector space. |
| $\text{abex}(K)$ | The set of absolute matrix extreme points of $K$. |
| $\text{co}(K)$ | The smallest matrix convex set containing $K$. |

# Abstract

In this work, we study operator systems through their matrix state spaces and matrix ranges. It is known that if $\mathbf{s} = (s_1, \ldots, s_d)$ and $\mathbf{r} = (r_1, \ldots, r_d)$ are two $d$-tuples of operators, then there exists a complete order isomorphism that maps $s_i$ to $r_i$ for $i = 1, \ldots, d$ if and only if their matrix ranges are equal. We show similar result hold for general operator systems. Namely the two operator systems are complete order isomorphic if and only if their matrix state spaces are matrix affine homeomorphic.

We then study the question, to what extent do the lower matrix levels encode the complete order structure of an operator system. In particular, we ask when can the complete order structure of an operator system can be recovered from the $N$-order structure for some finite $N$. We show that when the boundary representations of an operator system are of bounded dimension, this is indeed the case. We use this result to classify certain operator systems through their $N$-order structure.

Lastly, we consider a special class of operator systems generated by $q$-commuting pairs or $\Lambda$-commuting $d$-tuples of unitaries, and characterize their matrix state spaces /matrix ranges in terms of their boundary representations. We show that for the rational case, $q$ or $\Lambda$ will be an invariant of the $N$-order structure for some $N \in \mathbb{N}$ which depends on $q$ or $\Lambda$. In particular, we show that two representations of such tuples are equivalent if and only if they admit the same $N$-order structure.

# Chapter 1

# Introduction

Let $\mathcal{S}$ be an operator system and consider its matrix state space $\mathrm{UCP}(\mathcal{S})$ (see Chapter 2 for the full definition). For this compact matrix convex set, we can consider the set of matrix affine functions $A(\mathrm{UCP}(\mathcal{S}))$. It is due to Webster and Winkler (see [22]), that $\mathcal{S}$ can be identified in a natural way with $A(\mathrm{UCP}(\mathcal{S}))$, which means that we can recover $\mathcal{S}$ by studying the matrix convex structure of $\mathrm{UCP}(\mathcal{S})$. Conversely, for any compact matrix convex set $K$, the set $\mathrm{UCP}(A(K))$ can be identified in a natural way with $K$.

This shows us that one can use the tools related to matrix convexity to study operator systems, and vice versa. In particular, we are interested in classifying the complete order equivalence class of certain operator systems by considering matrix affine maps between their matrix state spaces.

In the first part of the work, we characterize compact matrix convex sets (in particular, matrix state spaces and matrix ranges) in terms of the matrix extreme points, by utilizing some results from the boundary theory (for example, [8],[11],[2]). Several characterizations of this sort already exist ([14],[9]), but they are formulated for self-adjoint tuples of matrices. The first part of Chapter 3 is dedicated to generalizing some of these results for general compact matrix convex sets. We then turn our attention to a special class of operator systems, namely, those with matrix state spaces which are finitely generated (see definitions in Chapter 2). We prove the following two theorems:

**Theorem 3.18.** *Let $\mathcal{S}, \mathcal{R}$ be operator systems such that $\mathrm{UCP}(\mathcal{S}), \mathrm{UCP}(\mathcal{R})$ are $N_{\mathcal{S}}$-generated and $N_{\mathcal{R}}$-generated for some $N_{\mathcal{S}}, N_{\mathcal{R}} \in \mathbb{N}$, and set $N = \max\{N_{\mathcal{S}}, N_{\mathcal{R}}\}$. Then, $\mathcal{S}$ and $\mathcal{R}$ are $N$-order isomorphic if and only if they are completely order isomorphic.*

**Theorem 3.19.** *Let $\mathcal{S}$ be an operator system such that $\mathrm{UCP}(\mathcal{S})$ is $N$-generated. Then, for any operator system $\mathcal{R}$, if $\phi : \mathcal{R} \to \mathcal{S}$ is unital and $N$-positive, it is UCP.*

The first theorem will show that being $N$-generated means that in order to classify $\mathcal{S}$ as an operator system, it is sufficient to consider the $N$-order structure instead of the complete order structure. The second theorem shows that operators with $N$-generated matrix state spaces satisfy an analog of Theorem 2.7. The rest of the chapter gives analogous results for matrix ranges of $d$-tuples which are finitely generated.

In Chapter 4 we discuss a special case of operator systems which are generated by a $d$-tuple of unitaries. In this case, we show in Corollary 4.5, that whenever two tuples are completely order equivalent, they generate isomorphic $C^*$-algebras. We also show

that in addition, when the tuples have finitely generated matrix ranges, then being $N$-order equivalent also means that they generated isomorphic $C^*$-algebras.

Finally, in Chapter 5, we consider a special case of $d$-tuples of unitaries, rotation operator systems generated by a $\Lambda$-commuting $d$-tuples or by a $q$-commuting pair of unitaries. We give a more concrete characterization of the matrix state space and matrix range of those systems, and use this to classify those operator systems up to complete order equivalence. We prove the following:

**Theorem 5.21.** *Let $\Lambda$ be rational and let $\mathbf{s}$ be a $d$-tuple of $\Lambda$-commuting unitaries. For a $d$-tuple $\mathbf{r}$ of $\tilde{\Lambda}$-commuting unitaries, the following are equivalent:*

1. *$\mathbf{s}$ and $\mathbf{r}$ are $N(\Lambda)$-order equivalent.*
2. *$\mathbf{s}$ and $\mathbf{r}$ are completely order equivalent.*
3. *$\mathcal{W}_{N(\Lambda)}(\mathbf{s}) = \mathcal{W}_{N(\Lambda)}(\mathbf{r})$.*
4. *$\mathcal{W}(\mathbf{s}) = \mathcal{W}(\mathbf{r})$.*
5. *$\mathbf{s}$ and $\mathbf{r}$ are $*$-isomorphic.*

In Section 5.4 we turn our attention to the problem of classifying rotation operator systems up to unital isometries. Using dilation theoretic methods (and result from [12]) to show that in some cases, the difference of two rotation systems can be detected at the first level of the matrix range, and we also give an example to show that this technique does not always work.

# Chapter 2

# Preliminaries and notation

## 2.1 Operator systems and positivity

Given a unital $C^*$-algebra $\mathcal{A}$, we say that $\mathcal{S} \subset A$ is an *operator system* if it contains the unit and it is closed under the $*$-operation. Given a tuple $\mathbf{s} = (s_1, \ldots, s_n)$ of elements in $\mathcal{A}$, we use the notation $\mathcal{S}(\mathbf{s})$ for the operator systems generated by $\mathbf{s}$.

We say that an element $s \in \mathcal{S}$ is *positive* if it is positive as an element of the $C^*$-algebra $\mathcal{A}$ containing $\mathcal{S}$. For two elements $s, s'$, we will say that $s \geq s'$ is $\mathcal{S}$ if $s - s'$ is positive.

For any operator system $\mathcal{S}$ and for all $n \in \mathbb{N}$, we will denote by $\mathbb{M}_n(\mathcal{S}) \subset \mathbb{M}_n(\mathcal{A})$ the $n \times n$ matrices with entries from $\mathcal{S}$. $\mathbb{M}_n(\mathcal{S})$ is a vector space with a $*$-operation given by $(s_{ij})^* = \left(s_{ji}^*\right)$, which makes it an operator system in the $C^*$-algebra $\mathbb{M}_n(\mathcal{A})$. The notion of positivity in $\mathbb{M}_n(\mathcal{S})$ is defined in the same way.

## 2.2 *-homomorphisms, positive maps and completely positive maps

For two $C^*$-algebras $\mathcal{A}, \mathcal{B}$, we say that $\pi : \mathcal{A} \to \mathcal{B}$ is a $*$-homomorphism if it is an algebra homomorphism which satisfies $\pi(a^*) = \pi(a)^*$ for all $a \in \mathcal{A}$. If $\pi$ is a bijection, we say that $\pi$ is a $*$-isomorphism.

When $\mathcal{S}$ is an operator system, we do not have a multiplicative structure in general and we want to discuss other mappings, which will be natural for the structure of operator systems. For this, note that if $\mathcal{S}$ is an operator system and $s \in \mathcal{S}$ is self-adjoint, then:

$$s = \frac{\|s\|\mathbb{1}_{\mathcal{S}} + s}{2} - \frac{\|s\|\mathbb{1}_{\mathcal{S}} - s}{2}$$

shows that $s$ is a linear combination of positive elements is $\mathcal{S}$. If $s$ is not self-adjoint, we can write it as:

$$s = \frac{s + s^*}{2} + i\frac{s - s^*}{2i}$$

which is a linear combination of two self-adjoint elements, and thus $s$ can be written as a linear combination of (at most 4) positive elements. Therefore, an operator system is generated by its positive elements. This motivates us to talk about mappings which

preserve both the linear structure and the positive cone (set of positive elements) of the operator systems.

**Definition 2.1.** Let $\mathcal{S}, \mathcal{R}$ be two operator systems and $\phi : \mathcal{S} \to \mathcal{R}$ be a linear map. We say that $\phi$ is:

- *Positive* if $\phi(s) \geq 0$ in $\mathcal{R}$ whenever $s \geq 0$ is $\mathcal{S}$.
- *n-positive* (for $n \in \mathbb{N}$) if the map $\phi_n : \mathbb{M}_n(\mathcal{S}) \to \mathbb{M}_n(\mathcal{R})$ given by $\phi(s_{ij}) = (\phi(s_{ij}))$ is positive.
- *Completely positive* if $\phi$ is $n$-positive for all $n \in \mathbb{N}$. We shall abbreviate and say that such a map is CP.
  - If a CP map satisfies $\phi(\mathbb{1}_{\mathcal{S}}) = \mathbb{1}_{\mathcal{R}}$, we say it is unital and completely positive. We will also abbreviate this and call such a map UCP.

Next, we define the kinds of isomorphisms we may consider between operator system.

**Definition 2.2.** Let $\mathcal{S}, \mathcal{R}$ be two operator systems and $\phi : \mathcal{S} \to \mathcal{R}$ be a linear bijection. We say that $\phi$ is an:

- *Order isomorphism* if $\phi$ is unital and both $\phi, \phi^{-1}$ are positive.
- *n-order isomorphism* if $\phi_n$ is an order isomorphism. Namely, if $\phi$ is unital and both $\phi, \phi^{-1}$ are $n$-positive.
- *Complete order isomorphism* if $\phi$ is an $n$-order isomorphism for all $n \in \mathbb{N}$. In other words, if $\phi, \phi^{-1}$ are both UCPs.

Aside from general operator systems, we will also deal with operator systems generated by some $d$-tuple $\mathbf{s}$. In this case, it will be interesting to discuss certain equivalence relation between two tuples through the operator systems that they generate.

**Definition 2.3.** Let $\mathbf{s}, \mathbf{r}$ be two $d$-tuples. We say that:

- $\mathbf{s}, \mathbf{r}$ are *n-order equivalent* if there exists an $n$-order isomorphism of $\mathcal{S}(\mathbf{s})$ and $\mathcal{S}(\mathbf{r})$ mapping $s_i$ to $r_i$ for all $i$.
- $\mathbf{s}, \mathbf{r}$ are *completely order equivalent* if they are $n$-order equivalent for all $n \in \mathbb{N}$.
- $\mathbf{s}, \mathbf{r}$ are $*$-*isomorphic* if the there is a $*$-isomorphism of $C^*(\mathbf{s})$ and $C^*(\mathbf{r})$ mapping $s_i$ to $r_i$ for all $i$.

There are two types of common completely positive maps that will appear throughout the text:

- If $\pi : \mathcal{A} \to \mathcal{B}$ is a unital $*$-homomorphism between the two $C^*$-algebras, then for any operator system $\mathcal{S}$ in $\mathcal{A}$, the map $\pi\big|_{\mathcal{S}}$ is UCP.
- Assume $\mathcal{A} \subset B(\mathcal{H})$ for some $\mathcal{H}$ and let $V \in B(\mathcal{K}, \mathcal{H})$. Then, the map $a \mapsto V^* a V$ is completely positive, and when $\mathcal{A}$ is unital and $V^*V = \mathbb{1}_{\mathcal{K}}$, it is UCP.

- In general, we know by [21, Theorem 1], that any completely positive map $\phi : \mathcal{A} \to B(\mathcal{H})$ where $\mathcal{A}$ is a unital $C^*$-algebra is of the form $\phi(a) = V^*\pi(a)V$ for a $*$-homomorphism $\pi : \mathcal{A} \to B(\mathcal{K})$ (for some Hilbert space $\mathcal{K}$, and a bounded operator $V : \mathcal{H} \to \mathcal{K}$). This shows that for $C^*$-algebras, the CP maps are all compositions of conjugations and $*$-homomorphisms.

The following are some basic results we will need later.

**Lemma 2.4.** *[20, Lemma 2.11] Let $\mathcal{S}$ be an operator system, $\mathcal{B}$ a unital $C^*$-algebra, and $\phi : \mathcal{S} \to \mathcal{B}$ a unital contraction. Then, $\phi$ is positive.*

*Proof.* Without loss of generality, we may assume that $\mathcal{B} = B(\mathcal{H})$ for some Hilbert space $\mathcal{H}$. Given any $\xi \in \mathcal{H}$ of norm 1, we set $f : \mathcal{S} \to \mathbb{C}$ by $f(s) = \langle \phi(s)\xi, \xi \rangle$. As shown in [20, Lemma 2.10], a contractive and unital functional is positive. $f$ is such a functional and thus positive, which means that $\langle \phi(s)\xi, \xi \rangle$ will be positive for all $\xi \in \mathcal{H}$, so $\phi(s)$ is positive. □

**Lemma 2.5.** *Let $\mathcal{S}, \mathcal{R}$ be two operator systems (contained in $C^*$-algebras $\mathcal{A}, \mathcal{B}$ respectively) and let $\phi : \mathcal{S} \to \mathcal{R}$ be $n$-positive. Then, $\phi$ is $k$-positive for all $k < n$.*

*Proof.* The map $\iota_{k,n} : \mathbb{M}_k(\mathcal{A}) \to \mathbb{M}_n(\mathcal{A})$ given by:

$$\iota_{k,n}(s_{ij}) = \begin{pmatrix} (s_{ij}) & 0 \\ 0 & 0 \end{pmatrix}$$

is a $*$-homomorphism and thus completely positive. The map $p_{n,k} : \mathbb{M}_n(\mathcal{B}) \to \mathbb{M}_k(\mathcal{B})$ given by:

$$p_{n,k}\Big((r_{ij})_{1 \le i,j \le n}\Big) = (r_{ij})_{1 \le i,j \le k}$$

can also be written as $p_{n,k}(r) = V^* r V$ where $V$ is the matrix:

$$V = \begin{pmatrix} \mathbb{1}_k \\ 0 \end{pmatrix}.$$

which means that $p_{n,k}$ is completely positive. Therefore, if $\phi$ is $n$ positive, then:

$$\phi_k = p_{n,k} \circ \phi_n \circ \iota_{k,n}$$

is a composition of positive maps and thus positive. □

**Lemma 2.6.** *Let $\mathcal{S}, \mathcal{R}$ be two operator systems (contained in $C^*$-algebras $\mathcal{A}, \mathcal{B}$ respectively) and let $\phi : \mathcal{S} \to \mathcal{R}$ be positive. Then $\phi$ is self-adjoint, in meaning $\phi(s^*) = \phi(s)^*$ for all $s \in \mathcal{S}$.*

*Proof.* Assume first that $s \in \mathcal{S}$ is self-adjoint. Then:

$$s = \frac{\|s\|\mathbb{1}_{\mathcal{S}} + s}{2} - \frac{\|s\|\mathbb{1}_{\mathcal{S}} - s}{2}$$

is a decomposition of $s$ to a difference of two positive elements is $\mathcal{S}$, and thus:

$$\phi(s) = \frac{1}{2}\phi(\|s\|\mathbb{1}_{\mathcal{S}} + s) - \frac{1}{2}\phi(\|s\|\mathbb{1}_{\mathcal{S}} - s)$$

is a decomposition of $\phi(s)$ to a difference of two positive elements in $\mathcal{R}$ and thus self-adjoint. For a general element $s$ we write $s = s_1 + is_2$ such that $s_1, s_2$ are self-adjoint, so:

$$\phi(s^*) = \phi(s_1 - is_2) = \phi(s_1) - i\phi(s_2) = (\phi(s_1) + i\phi(s_2))^* = \phi(s)^*$$

which completes the proof. □

It is obvious that $*$-homomorphisms are unital and completely positive, and that unital and completely positive is $n$ positive for all $n$. It is not true in general that an $n$-positive map is completely positive and a completely positive map need not extend to a $*$-homomorphism of the generated $C^*$-algebra. In Chapters 3,4 and 5 we develop tools to determine some cases for which an $n$-positive map will automatically be completely positive, and some cases for which a completely positive map will automatically extend to a $*$-homomorphism of the generated $C^*$-algebra.

## 2.3 Matrix convex sets

As stated in the previous section, it would be interesting to see when $n$-positivity implies complete positivity. The following theorem give us a large family of examples for such cases.

**Theorem 2.7.** *[4, Theorem 5] Let $\mathcal{A}$ be a unital $C^*$-algebra, and let $\phi : \mathcal{A} \to \mathbb{M}_n$ be $n$-positive. Then, $\phi$ is completely positive.*

In fact, looking at the proof shows that the theorem is true even for maps from operator systems to $\mathbb{M}_n$. In particular, we have that two $d$-tuples in $\mathbb{M}_n$ and $\mathbb{M}_m$ are $\max\{n, m\}$-order equivalent if and only if they are completely order equivalent.

For the case of two general operator systems $\mathcal{S}, \mathcal{R}$, this theorem does not state anything about maps from $\mathcal{S}$ to $\mathcal{R}$, but it does give us some tools to study operator systems. Namely, by considering maps from $\mathcal{S}$ to $\mathbb{M}_n$ for some $n$. In Chapter 3 we see that these maps already encode the $n$ order (and in some cases, the complete order) structure of the operator system. We would like, then, to consider the set of all maps from $\mathcal{S} \to \mathbb{M}_n$ which are $n$-positive (or completely positive, since this is equivalent) for all $n \in \mathbb{N}$.

It turns out that this set has a very useful structure endowed on it, which makes it a *compact matrix convex set*. We remind the reader of the main notions and results we will need in this work.

**Definition 2.8.** A *matrix convex set* in a vector space $V$, is a collection $K = (K_n)_{n\in\mathbb{N}}$ of subsets $K_n \subset \mathbb{M}_n(V)$ such that:

$$\sum_{i=1}^{k} \gamma_i^* v_i \gamma_i \in K_n$$

for all $v_i \in K_{n_i}$ and $\gamma_i \in \mathbb{M}_{n_i,n}$ for $i = 1, \ldots, k$ satisfying $\sum_i \gamma_i^* \gamma_i = \mathbb{1}_n$.

We say that $v = \sum_i \gamma_i^* v_i \gamma_i$ is a *matrix convex combination* of $v_1, \ldots, v_k$.

If $V$ is a locally convex vector space, we can also say that $K = (K_n)_{n\in\mathbb{N}}$ is a *compact matrix convex set* if each $K_n$ is compact in $M_n(V)$ in the product topology.

Below are some examples of matrix convex sets which we will use later on.

**Definition 2.9.** Let $\mathcal{S}$ be an operator system. The *matrix state space* is the union $\mathrm{UCP}(\mathcal{S}) = (\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n))_{n=1}^{\infty}$ where:

$$\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n) := \{\phi : \mathcal{S} \to \mathbb{M}_n | \phi \text{ is UCP}\}.$$

Matrix convex combinations in this set are defined pointwise. Namely:

$$\left(\sum_i \gamma_i^* \varphi_i \gamma_i\right)(s) = \sum_i \gamma_i^* \phi_i(s) \gamma_i.$$

Note that $\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$ may be identified with a convex subset of $\mathbb{M}_n(\mathcal{S}^*)$, which means we can endow it with the product topology from $\mathcal{S}^*$, which is the weak* topology from $\mathcal{S}$. Therefore we say that $\mathrm{UCP}(\mathcal{S})$ is matrix convex in $\mathcal{S}^*$.

The term *matrix states* generalizes the term *state* of an operator system, which refers to a unital and completely positive functional (in our case, those are unital and completely positive maps into $\mathbb{M}_n$ instead of $\mathbb{C}$).

Note that $\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$ is a closed subset of $\mathrm{CP}(\mathcal{S}, \mathbb{M}_n) \cap B_1(\mathcal{S}, \mathbb{M}_n)$ (namely, the set of all completely positive maps bounded by 1), and the latter was proved to be compact (see [2]) and therefore, $\mathrm{UCP}(\mathcal{S})$ is a compact matrix convex set.

**Definition 2.10.** Let $\mathcal{S} = \mathcal{S}(\mathbf{s})$ be an operator system generated by a $d$-tuple $\mathbf{s}$. The *matrix range* of $\mathcal{S}$ is the collection $\mathcal{W}(\mathbf{s}) = (\mathcal{W}_n(\mathbf{s}))_{n=1}^{\infty}$ where:

$$\mathcal{W}_n(\mathbf{s}) = \{(\phi(s_1), \ldots, \phi(s_d)) | \phi \in \mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)\} \subset \mathbb{M}_n^d, \quad \forall n \in \mathbb{N}$$

Much can be deduced about operator systems by studying their matrix ranges. For example (which will be widely used later on) is the following theorem by Davidson, Dor-On, Shalit and Solel:

**Theorem 2.11.** *[7, Theorem 5.1] Let $\mathbf{s} \in B(\mathcal{H})^d, \mathbf{r} \in B(\mathcal{K})^d$ be $d$-tuples of operators.*

1. *Given $n \in \mathbb{N}$, if there exists a unital $n$-positive map $\phi : \mathcal{S}(\mathbf{s}) \to \mathcal{S}(\mathbf{r})$ sending $\mathbf{s}$ to $\mathbf{r}$, then $\mathcal{W}_n(\mathbf{r}) \subset \mathcal{W}_n(\mathbf{s})$.*
2. *There exists a UCP map $\phi : \mathcal{S}(\mathbf{s}) \to \mathcal{S}(\mathbf{r})$ sending $\mathbf{s}$ to $\mathbf{r}$ if and only if $\mathcal{W}(\mathbf{r}) \subset \mathcal{W}(\mathbf{s})$. If $\mathbf{r}$ is a commuting tuple of normal operators, then this inclusion is equivalent to $\sigma(\mathbf{r}) \subset \mathcal{W}_1(\mathbf{s})$ which is also equivalent to $\mathcal{W}_1(\mathbf{r}) \subset \mathcal{W}_1(\mathbf{s})$.*
3. *There exists a unital completely isometric map $\phi : \mathcal{S}(\mathbf{s}) \to \mathcal{S}(\mathbf{r})$ sending $\mathbf{s}$ to $\mathbf{r}$ if and only if $\mathcal{W}(\mathbf{s}) = \mathcal{W}(\mathbf{r})$.*

Note that the values $\phi(s_i)$ for $i = 1, \ldots, d$ determine $\phi$ uniquely (also follows from Lemma 2.6), which gives rise to a correspondence between $\mathrm{UCP}(\mathcal{S})$ and $\mathcal{W}(\mathbf{s})$. This correspondence is given by a *matrix affine homeomorphism* which will be defined later on, and will give a proof as to why the matrix range is also a compact matrix convex set.

It is worth noting that matrix ranges are the typical form of compact matrix convex sets in $\mathbb{M}^d$. Namely, any compact matrix convex set is of the form $\mathcal{W}(\mathbf{s})$, as shown in [15, Proposition 31]. Therefore, studying matrix ranges (or, from another perspective, the UCP maps of the corresponding operator system), might also shed light on the theory of compact matrix convex sets.

**Definition 2.12.** Let $\mathcal{C} = (C_n)_{n=1}^{\infty}$ be a closed and bounded matrix convex set in $\mathbb{C}^d$ (namely, $C_n \subset \mathbb{M}_n^d$ for all $n \in \mathbb{N}$). We set:

$$\mathcal{W}^{k\text{-min}}(\mathcal{C}) := \bigcap \left\{ \mathcal{L} = \bigcup_{n\in\mathbb{N}} L_n \middle| \mathcal{L} \text{ is matrix convex such that } L_k = C_k \right\}.$$

One can think of such set as the minimal matrix convex set that has $C_k$ as its $k$-th level.

It is also possible to consider sets of the form:

$$\mathcal{W}^{\leq k\text{-min}}(\mathcal{C}) := \bigcap \left\{ \mathcal{L} = \bigcup_{n\in\mathbb{N}} L_n \middle| \mathcal{L} \text{ is matrix convex such that } L_m = C_m \ \forall 1 \leq m \leq k \right\}.$$

Note that by definition, $\mathcal{W}^{\leq k\text{-min}}(\mathcal{C}) \subset \mathcal{W}^{k\text{-min}}(\mathcal{C})$. But in the next section we show that those sets are in fact equal, so for the rest of the work we use the former notation, $\mathcal{W}^{k\text{-min}}(\mathcal{C})$.

When $k = 1$ and $C \subset \mathbb{C}^d$ is a bounded, closed and convex set, we set:

$$\mathcal{W}^{\min}(C) := \bigcap \left\{ \mathcal{L} = \bigcup_{n\in\mathbb{N}} L_n \middle| \mathcal{L} \text{ is matrix convex such that } L_1 = C \right\}.$$

There is a similar notion of maximal matrix convex sets:

$$\mathcal{W}^{k\text{-max}}(\mathcal{C}) = \bigcup_{n\in\mathbb{N}} \left\{ \mathbf{s} \in \mathbb{M}_n^d \middle| \mathcal{W}_k(\mathbf{s}) \subset C_k \right\},$$

which is easily verified to be a matrix convex set which contains any matrix convex set with $C_k$ at it's $k$th level. In particular, if $\mathcal{L}$ is a matrix convex set such that $L_k = C_k$, then:

$$\mathcal{W}^{k\text{-min}}(\mathcal{C}) \subset \mathcal{L} \subset \mathcal{W}^{k\text{-max}}(\mathcal{C}).$$

The definitions of $k$-minimal and $k$-maximal sets were introduced in a work by Paulson and Passer (see [18] with references to earlier works). In a sense (which will become clearer in Chapter 3 and Chapter 5), $k$-minimal matrix convex sets are matrix convex sets for which all of the information about the matrix convex set lie within finitely many levels.

When $\mathcal{C} = \mathcal{W}(\mathbf{s})$ for some tuple $\mathbf{s}$, we will use the notations:

$$\mathcal{W}^{k\text{-min}}(\mathbf{s}) := \mathcal{W}^{k\text{-min}}(\mathcal{W}(\mathbf{s}))$$

for convenience. These matrix convex sets appear naturally in the study of certain UCP interpolation problems (which will appear in this study as well, although not explicitly), see [19] and [18] for examples.

### 2.3.1 Matrix extreme points

Recall that if $V$ is a vector space and $K \subset V$ is convex, a point $v \in K$ is said to be *extreme* if whenever $v = (1-\lambda)u + \lambda w$ for some $u, w \in K$ and $\lambda \in (0,1)$, then $u = w = v$.

The Krein-Milman theorem shows us that the extreme points of a compact convex set of a locally convex vector space recover the set completely, so that:

$$K = \overline{\text{conv}(\text{ext}(K))}.$$

A matrix convex set is a generalization of a convex set, in the sense that any convex set admits a matrix convex set with the first level being that set. Therefore, one would hope to be able to generalize the Krein-Milman theorem for matrix convex sets, which means we first need to be able to generalize the notion of extremality.

**Definition 2.13.** If $K = (K_n)_{n\in\mathbb{N}}$ is a matrix convex set in a vector space $V$, we say that $v = \sum_i \gamma_i^* v_i \gamma_i$ is a *proper* matrix convex combination, if it is a matrix convex combination and $\gamma_i$ is surjective for all $i$.

Note that $\gamma_i$ being surjective means that $n \geq n_i$. One may consider proper matrix convex combinations as combinations "from below".

The following definition is a generalization of an extreme point, which is due to Webster and Winkler (see [22]).

**Definition 2.14.** Let $K = (K_n)_{n\in\mathbb{N}}$ be a matrix convex set. We say that $v \in K_n$ is a *matrix extreme point* if whenever $v$ is a proper matrix convex combination $v = \sum_{i=1}^k \gamma_i^* v_i \gamma_i$, then for all $i = 1, \ldots, k$, $n_i = n$ and $v = u_i^* v_i u_i$ for some unitary $u_i \in \mathbb{M}_n$.

In simpler terms, a point is matrix extreme if it cannot be written as a matrix convex combination from lower levels except in some trivial way (so that even when the summands are from the same level, they are unitarily equivalent to that point).

**Definition 2.15.** Let $K = (K_n)_{n\in\mathbb{N}}$ be a matrix convex set. For all $n \in \mathbb{N}$, we set $\partial K_n$ as the set of all matrix extreme points of $K$ in $K_n$. The set $\partial K = (\partial K_n)_{n\in\mathbb{N}}$ is then the set of all matrix extreme points of $K$.

Note that an extreme point in level $K_1$ must also be a matrix extreme point, because the only proper matrix convex combinations which are in $K_1$ are those with summands from $K_1$. In that case, the matrices $\gamma_i$ are all scalars which makes this a euclidean convex combination, which is trivial when the point is an extreme point.

In Chapter 3 we will show that the converse is true for any level. Namely, if $v \in K_n$ is matrix extreme, it is also an extreme point in $K_n$.

We now refer to a stronger notion of extremality (first introduced by Evert-Helton-Klep-McCullough (see [9]), which will turn out to be essential for some calculations in Chapter 5.

**Definition 2.16.** Let $K = (K_n)_{n\in\mathbb{N}}$ be matrix convex. We say that $v \in K_n$ is an *absolute matrix extreme point* if whenever $v$ is a matrix convex combination (not necessarily proper) of $v_i \in K_{n_i}$ for $i = 1, \ldots, k$, then for all $i$, if $n_i \leq n$, then $v = u_i^* v_i u_i$ for some unitary $u_i \in \mathbb{M}_n$, and if $n_i > n$, then $v_i = u_i^*(v \oplus c_i)u_i$ for some unitary $u_i \in \mathbb{M}_{n_i}$ and $c_i \in K$. We denote by $\text{abex}(K)$ the set of all absolute matrix extreme points of $K$.

In other words, an absolute matrix extreme point cannot be written as a matrix convex combination from any level except in a trivial way.

Absolute matrix extreme points are always matrix extreme points, but the converse is not always true, as the next example shows.

**Example 2.17.** Let $F_1, F_2$ be the unitary anti-commuting matrices given by:

$$F_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad F_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Using Theorem 5.6, we get that $\mathcal{W}(F_1, F_2)$ has absolute matrix extreme points only at level 2, and in Example 3.22, we prove that $\mathcal{W}_1(F_1, F_2) = \overline{\mathbb{B}}_2$ (the closed unit ball in $\mathbb{R}^2$). As we stated before, an extreme point in $\mathcal{W}_1(F_1, F_2)$ must be matrix extreme, so any point in the unit circle (being an extreme point of the ball) must be a matrix extreme point which is not absolute.

As we will see later on, any compact matrix convex set has matrix extreme points but there are matrix convex sets which have no absolute extreme points. For this reason, we cannot hope for absolute matrix extreme points to generalize the Krein-Milman theorem, but luckily the matrix extreme points which are not necessarily absolute, do give this generalization, which is due to Webster and Winkler.

**Definition 2.18.** Let $V$ be a vector space and let $S = (S_n)_{n=1}^{\infty}$ be a non-empty subset of $(\mathbb{M}_n(V))_{n=1}^{\infty}$. The *matrix convex hull* of $S$ is defined as the smallest matrix convex set containing $S$. More explicitly:

$$\mathrm{co}(S) = \left\{ \sum_i \gamma_i^* v_i \gamma_i \,\middle|\, \gamma_i \in \mathbb{M}_{n_i,n},\ v_i \in S_i \text{ for all } i \text{ and } \sum_i \gamma_i^* \gamma_i = \mathbb{1}_n \right\}$$

**Theorem 2.19.** *[22, Thoerem 4.3] Let $K = (K_n)_{n=1}^{\infty}$ be a compact matrix convex set in a locally convex space $V$, and let $\partial K = (\partial K_n)_{n=1}^{\infty}$ denote the collection of matrix extreme points $\partial K_n$ of $K_n$. Then, $\partial K$ is non-empty, and:*

$$K = \overline{\mathrm{co}}(\partial K)$$

Thus, matrix extreme points are the generalization of extreme points in locally convex sets, in the sense that they also share certain extremal properties (but with matrix convex combinations which reduce to convex combinations at level 1), and in the sense that matrix extreme points are always extreme points in their level.

This means that by studying $\partial\mathrm{UCP}(\mathcal{S})$ (or $\partial\mathcal{W}(\mathbf{s})$), we can extract information about $\mathrm{UCP}(\mathcal{S})$ (or $\mathcal{W}(\mathbf{s})$), and then use this perspective to study $\mathcal{S}$. Therefore, the next goal is to characterize matrix extreme points and absolute matrix extreme points in such sets. As we will show in Chapter 3, those points are linked to certain representations of the operator systems.

A special case of a compact matrix convex set is one which has an upper bound on the levels that contain a matrix extreme point.

**Definition 2.20.** Let $K = (K_n)_{n=1}^{\infty}$ be a compact matrix convex set. We say that $K$ is *$N$-generated*, if $\partial K_N \neq \emptyset$ and $\partial K_n = \emptyset$ for all $n > N$.

The Webster-Winkler Krein-Milman theorem shows that $N$-generated compact matrix convex sets can be recovered from their first $N$ levels. Applying this idea to $\mathrm{UCP}(\mathcal{S})$ and $\mathcal{W}(\mathbf{s})$ yields interesting results in chapter 3.

### 2.3.2 Matrix affine functions

Continuing the motivation for studying maps between two operator systems, it is natural to consider maps between some associated matrix convex sets (for example, maps between $\mathrm{UCP}(\mathcal{S})$ and $\mathrm{UCP}(\mathcal{R})$). Indeed, There is a natural notion of a morphism between matrix convex sets which is due to Webster and Winkler (see [22, Definition 3.4]).

**Definition 2.21.** Let $V, W$ be vector spaces and let $K = (K_n)_{n\in\mathbb{N}}, C = (C_n)_{n\in\mathbb{N}}$ be matrix convex sets over $V$ and $W$ respectively. A sequence $\boldsymbol{\theta} = (\theta_n)_{n\in\mathbb{N}}$, of mappings $\theta_n : K_n \to C_n$ is called a *matrix affine map* if:

$$\theta_n\left(\sum_{i=1}^{k}\gamma_i^* v_i \gamma_i\right) = \sum_{i=1}^{k}\gamma_i^*\theta_{n_i}(v_i)\gamma_i$$

for all $v_i \in K_{n_i}, \gamma_i \in \mathbb{M}_{n_i,n}$ for $i = 1,\ldots,k$ satisfying $\sum_i \gamma_i^*\gamma_i = \mathbb{1}_n$. If $V, W$ are locally convex vector spaces, we say that $\boldsymbol{\theta}$ is a *matrix affine homeomorphism* if for all $n \in \mathbb{N}$, $\theta_n$ is a homeomorphism.

If $v \in K_n$ is a matrix extreme point/absolute extreme point, it is clear that whenever $\boldsymbol{\theta}$ is a matrix affine homeomorphism, $\theta_n(v)$ is a matrix extreme point/absolute matrix extreme point.

**Example 2.22.** Let $\mathcal{S} = \mathcal{S}(\mathbf{s})$ be an operator system generated by a tuple $\mathbf{s}$. Define the sequence $\boldsymbol{\theta} := (\theta_n)_{n\in\mathbb{N}}$ of mappings $\theta_n : \mathrm{UCP}(\mathcal{S}, \mathbb{M}_n) \to \mathcal{W}_n(\mathbf{s})$ by:

$$\theta_n(\phi) = (\phi(s_1),\ldots,\phi(s_d)).$$

Note that by the definition of $\mathcal{W}_n(\mathbf{s})$, $\theta_n$ is surjective for all $n$. The fact that $\theta_n$ is injective follows from that fact that $\phi(s_1),\ldots,\phi(s_d)$ determine $\phi$ uniquely (recall Lemma 2.6, which says that as a completely positive map, $\phi(s^*) = \phi(s)^*$). As for continuity, note that the topology on $\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$ is the weak* topology induced by $\mathcal{S}$. Therefore, if $\phi_m$ converges to $\phi$, then $\phi_m(s_i)$ converges to $\phi(s_i)$ which means that $\theta_n(\phi_m)$ converges to $\theta_n(\phi)$. Since $\mathcal{W}_n(\mathbf{s})$ is Hausdorff and $\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$ is compact, $\theta_n$ being continuous means it is a homeomorphism.

As a consequence we get that there is a one-to-one correspondence between matrix/absolute extreme points in $\mathrm{UCP}(\mathcal{S})$ and $\mathcal{W}(\mathbf{s})$. The advantage of the correspondence is that in many cases, it is more convenient to work in one setting over the other. Obviously, $\mathrm{UCP}(\mathcal{S})$ is more general because it does not depend on the choice of a specific tuple of elements. But when we are considering an interesting tuple which generates $\mathcal{S}$, working through the setting of the matrix range opens us to many other useful tools which are formulated in that language.

**Definition 2.23.** Let $K = (K_n)_{n\in\mathbb{N}}$ be a compact matrix convex set. We denote by $A(K, \mathbb{C})$ (or just $A(K)$) the set of all sequences $F = (F_n)_{n=1}^{\infty}$ of mappings $F_n : K_n \to \mathbb{M}_n$ such that $F$ is matrix affine and $F_1$ is continuous.

We bring here the construction of Webster and Winkler [22], which makes $A(K)$ into an abstract operator system, and then use this fact to show that $\mathcal{S}$ and $A(\mathrm{UCP}(\mathcal{S}))$ are completely order isomorphic.

We first consider $A(K, \mathbb{M}_r)$ which is the set of matrix affine mappings $F = (F_n)_{n=1}^{\infty}$ such that $F_n : K_n \to \mathbb{M}_n(\mathbb{M}_r)$ and $F_1$ is continuous. By considering the usual linear structure and adjoint operations on $\mathbb{M}_n(\mathbb{M}_r)$, we have that $A(K, \mathbb{M}_r)$ can be endowed with a vector space structure with a $*$-operation given by:

$$F_n^*(v) := (F_n(v))^*$$

for all $n \in \mathbb{N}, v \in K_n$. Moreover, we can define a positive cone in $A(K, \mathbb{M}_r)$ where $F = (F_n)_{n=1}^{\infty}$ is positive if $F_n(v) \geq 0$ for all $n \in \mathbb{N}, v \in K_n$.

We now identify $\mathbb{M}_r(A(K))$ with $A(K, \mathbb{M}_r)$. In order to do this we first assume that $F$ is self-adjoint in $A(K, \mathbb{M}_r)$, so that for all $n$ and for all $v$, the values of $F_n(v)$ are uniquely defined by:

$$\xi^* F_n(v) \xi = \sum_{ij} \bar{\xi}_i \xi_j (F_n(v))_{ij} = F_1(\xi^* v \xi) = F_1\left( \sum_{ij} \bar{\xi}_i \xi_j v_{ij} \right)$$

where $\xi \in \mathbb{C}^n$ is a unit vector. Therefore, for each $(F_{ij}) \in \mathbb{M}_r(A(K))$, we can define $G = (G_n)_{n=1}^{\infty} \in A(K, \mathbb{M}_r)$ by defining:

$$G_1(v) = (F_{ij}(v)) \in \mathbb{M}_r$$

and extending it (first for self-adjoint matrices and then for all matrices). This map is injective, which will alow us to use the ordering on $A(K, \mathbb{M}_r)$ for $\mathbb{M}_r(A(K))$.

Lastly, choosing $I = (I_n)_{n=1}^{\infty} \in A(K)$ such that $I_n(v) = \mathbb{1}_n$ for all $n \in \mathbb{N}, v \in K_n$, makes $A(K)$ into a matrix ordered vector space with a $*$-operation and an archimedean matrix order unit $I$. Thus, $A(K)$ satisfies the Choi-Effros axioms (see [5]) for an abstract operator system.

**Proposition 2.24.** *[22, Proposition 3.5] Let $\mathcal{S}$ be an operator system and let $\mathrm{UCP}(\mathcal{S})$ be its matrix state space. Then $\mathcal{S}$ and $A(\mathrm{UCP}(\mathcal{S}))$ are completely order isomorphic.*

*Proof.* Define the linear map $\delta_{\mathcal{S}} : \mathcal{S} \to A(\mathrm{UCP}(\mathcal{S}))$ by:

$$(\delta_{\mathcal{S}}(s))(\phi) := \phi(s)$$

for all $s \in \mathcal{S}, \phi \in \mathrm{UCP}(\mathcal{S})$. For all $n \in \mathbb{N}$ we use our identification $\mathbb{M}_n(A(\mathrm{UCP}(\mathcal{S})))$ with $A(\mathrm{UCP}(\mathcal{S}), \mathbb{M}_n)$ so that $(\delta_{\mathcal{S}}(s))_n : \mathrm{UCP}(\mathcal{S}, \mathbb{M}_n) \to \mathbb{M}_n$ given by:

$$(\delta_{\mathcal{S}}(s))_n(\varphi) = \varphi(s)$$

is positive because $s \geq 0$ implies that $\varphi(s)$ is positive for any positive map. Next, note that by [1, Section II 1] $\{F_1 | F = (F_n)_{n=1}^{\infty} \in A(\mathrm{UCP}(\mathcal{S}))\}$ is isomorphic to $\mathcal{S}$ as function systems via the same mapping $\delta_{\mathcal{S}}$, which means it is also a bijection.

Lastly, the fact that $\delta_{\mathcal{S}}$ is a complete order isomorphism, follows from the fact that $s \geq 0$ is positive if and only if $\varphi(s)$ is positive for all $\varphi \in \mathrm{UCP}(\mathcal{S})$ (see the end of the proof in [5, Theorem 4.4]). □

# Chapter 3

# $N$-generated compact matrix convex sets

## 3.1 Characterization of matrix extreme points

As it turns out, there is a useful way to describe matrix extreme and absolute matrix extreme points when the sets we consider are matrix ranges or matrix state spaces. Before stating those results, we will need the following definitions of *pure* maps, *maximal* maps, and *boundary representations*.

**Definition 3.1.** Let $\mathcal{S}$ be an operator system and let $\varphi \in \mathrm{UCP}(\mathcal{S}, B(\mathcal{H}))$. We say that $\varphi$ is a *pure* map if whenever $\psi : \mathcal{S} \to B(\mathcal{H})$ is a CP map such that $\varphi - \psi$ is CP, there exists some $\lambda \in [0,1]$ such that $\psi = \lambda\varphi$.

**Definition 3.2.** Let $\mathcal{S}$ be an operator system and let $\varphi \in \mathrm{UCP}(\mathcal{S}, B(\mathcal{H}))$. We say that $\phi$ is *maximal* if whenever we have some $\psi \in \mathrm{UCP}(\mathcal{S}, B(\mathcal{K}))$ and an isometry $V : \mathcal{H} \to \mathcal{K}$ such that $\varphi(s) = V^*\psi(s)V$, then there is a UCP map $\rho : \mathcal{S} \to (V\mathcal{H})^\perp$ such that $\psi = (V\varphi V^*) \oplus \rho$ with respect to the decomposition $\mathcal{K} = V\mathcal{H} \oplus (V\mathcal{H})^\perp$.

A map $\psi$ satisfying $\varphi = V^*\psi V$ for some isometry $V$ is called a *dilation* of $\varphi$. We also define maximality at a pair in $\mathcal{S} \times \mathcal{H}$.

**Definition 3.3.** Let $\mathcal{S}$ be an operator system and let $\varphi \in \mathrm{UCP}(\mathcal{S}, B(\mathcal{H}))$. For $(s,h) \in \mathcal{S} \times \mathcal{H}$, we say that $\varphi$ is *maximal* at $(s,h)$ if whenever $\psi \in \mathrm{UCP}(\mathcal{S}, B(\mathcal{K}))$ is a dilation of $\varphi$, we get that:

$$V\varphi(s)h = \psi(s)Vh$$

From the definition it is straight forward to check that $\varphi$ is maximal precisely when it is maximal at every $(s,h) \in \mathcal{S} \times \mathcal{H}$.

Also note that since $V$ is an isometry, by identifying $\mathcal{H}$ with $V\mathcal{H}$ as a closed subspace of $\mathcal{K}$, we say that $\varphi$ is maximal at a point $(s,h)$, if whenever $\varphi = P_{\mathcal{H}}\,\psi|_{\mathcal{H}}$, then $\varphi(s)h = \psi(s)h$, and it is maximal if whenever we have such a dilation, then $\psi = \varphi \oplus \rho$ for some UCP $\rho \in \mathrm{UCP}(\mathcal{S}, B(\mathcal{K} \ominus \mathcal{H}))$.

**Definition 3.4.** Let $\mathcal{S}$ be an operator system in a unital $C^*$-algebra $\mathcal{A}$. We say that a representation $\pi : \mathcal{A} \to B(\mathcal{H})$ is a *boundary representation* for $\mathcal{S}$ if it is irreducible and the only UCP map $\mathcal{A} \to B(\mathcal{H})$ which extends $\pi\big|_{\mathcal{S}}$ is $\pi$.

This last definition was introduced by Arveson (see [2, Definition 2.1.1]), although it was not yet clear whether every operator system admits a boundary representation. This was finally proved by Davidson and Kennedy in [8, Theorem 3.1] (although Arveson did prove earlier for separable operator systems).

The first theorem which characterizes matrix extreme points in $\mathrm{UCP}(\mathcal{S})$, is due to Farenick:

**Theorem 3.5.** *[11, Theorem B] Let $\mathcal{S}$ be an operator system in a unital $C^*$-algebra $\mathcal{A}$.*

1. *A matrix state $\varphi$ on $\mathcal{S}$ is a matrix extreme point of the matrix state space $\mathrm{UCP}(\mathcal{S})$ if and only if $\varphi$ is pure.*
2. *Every pure map $\varphi \in \mathrm{UCP}(\mathcal{S})$ extends to a pure map $\Phi \in \mathrm{UCP}(\mathcal{A})$.*

This characterization gives us some examples of matrix extreme points, which are the finite dimensional boundary representations. In [2], Arveson proved that for $C^*$-algebras, the pure and completely positive maps from $\mathcal{A}$ to $B(\mathcal{H})$, are of the form $V^*\pi(\cdot)V$ where $\pi$ is an irreducible representation on some Hilbert space $\mathcal{K}$, and $V : \mathcal{H} \to \mathcal{K}$ is non zero. It is not immediate that the statement will hold for operator systems. This was finally achieved by Farenick and Tessier:

**Proposition 3.6.** *[10, Proposition 2.12] Let $\mathcal{A}$ be a unital $C^*$-algebra generated by an operator system $\mathcal{S}$, and let $\pi : \mathcal{A} \to B(\mathcal{K})$ be a boundary representation for $\mathcal{S}$. Then, $\pi\big|_{\mathcal{S}}$ is pure.*

Following this result, we have that whenever $\varphi \in \mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$ extends to a boundary representation for $\mathcal{S}$, it is a matrix extreme point. A somewhat more surprising fact is that boundary representations are absolute matrix extreme points. For this we need another property of boundary points, which is maximality, which comes as a consequence of [16, Theorem 1.2]. From this, we are ready to give a precise characterization of absolute matrix extreme points of $\mathrm{UCP}(\mathcal{S})$. This is a somewhat different formulation then the one proved by Evert-Helton-Klep-McCullough (see [9, Theorem 1.1]), which is more compatible with this work.

**Theorem 3.7.** *Let $\mathcal{S}$ be an operator system and let $\varphi \in \mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$. Then, $\varphi$ is an absolute matrix extreme point of $\mathrm{UCP}(\mathcal{S})$ if and only if $\varphi$ extends to a boundary representation on $C^*(\mathcal{S})$.*

*Proof.* Assume that $\varphi$ is a boundary representation and assume that

$$\varphi = \sum_i \gamma_i^* \varphi_i \gamma_i$$

is a matrix convex combination with $\gamma_i$ non-zero for all $i$. Note that $\varphi - \gamma_i^*\varphi_i\gamma_i$ is a completely positive map for all $i$, which means that by the purity of $\varphi$, we can find some $t_i \in [0,1]$ such that $\gamma_i^*\varphi_i\gamma_i = t_i\varphi$ so:

$$\left(t_i^{-\frac{1}{2}}\gamma_i\right)^* \varphi_i \left(t_i^{-\frac{1}{2}}\gamma_i\right) = \varphi.$$

From this it follows that $\left(t_i^{-\frac{1}{2}}\gamma_i\right)^* \left(t_i^{-\frac{1}{2}}\gamma_i\right) = \mathbb{1}_n$, and therefore, whenever $n_i \leq n$, it must hold that $n_i = n$ from which it follows that $t_i^{-\frac{1}{2}}\gamma_i$ is unitary. When $n_i > n$, we

get that $\varphi_i$ is a dilation of $\varphi$, which means that because boundary representations are maximal, $\varphi_i$ must be unitarily equivalent to $\varphi \oplus \rho_i$ for some $\rho_i \in \mathrm{UCP}(\mathcal{S}, n_i - n)$. This proves that $\varphi$ is an absolute matrix extreme point.

Conversely, assume that $\varphi$ is an absolute matrix extreme points, which means that it is pure. By [8, Lemma 2.1], a pure and maximal map extends to a boundary representation, so we need to show that $\varphi$ is maximal. Assuming that it is not maximal, we can apply Lemma 3.8, and find some matrix extreme point $\psi \in \mathrm{UCP}(\mathcal{S}, \mathbb{M}_{n+1})$, such that $\varphi = V^* \psi V$. $\varphi$ being an absolute matrix extreme point means that $\psi = U^*(\varphi \oplus \rho)U$ for some $\rho \in \mathrm{UCP}(\mathcal{S}, \mathbb{C})$. But this means that $\psi$ is a matrix convex combination of $\varphi$ and $\rho$, which can be achieved by choosing:

$$\gamma_1 = \mathbb{1}_n \oplus 0 \in \mathbb{M}_{n,n_i}, \quad \gamma_2 = 0 \oplus \mathbb{1}_{n-n_i} \in \mathbb{M}_{n-n_i,n_i}$$

so that we can write:

$$\psi = U^*(\gamma_1^* \varphi \gamma_1 + \gamma_2^* \rho \gamma_2)U = (\gamma_1 U)^* \varphi(\gamma_1 U) + (\gamma_2 U)^* \rho(\gamma_2 U)$$

and:

$$(\gamma_1 U)^*(\gamma_1 U) + (\gamma_2 U)^*(\gamma_2 U) = U^*(\gamma_1^* \gamma_1 + \gamma_2^* \gamma_2)U = U^* \mathbb{1}_n U = \mathbb{1}_n.$$

This contradicts $\psi$ being matrix extreme. Therefore, $\varphi$ must be maximal and thus extends to a boundary representation for $\mathcal{S}$. □

Maps in $\mathrm{UCP}(\mathcal{S})$ which have a unique UCP extension to a $*$-homomorphism of the generated $C^*$-algebra, are said to have the *unique extension property*. Our proof shows that a UCP map is an absolute matrix extreme point of $\mathrm{UCP}(\mathcal{S})$ if and only if it has this property (because this is equivalent to the map being a restriction of a boundary representation.

Note that we can characterize matrix state spaces which admit no absolute matrix extreme points. Those correspond to operator systems which admit no finite dimensional boundary representations. In Chapter 5 we will give concrete examples of such sets.

The lemma given below was used in the proof of Theorem 3.7, but it serves by itself as a tool to understand matrix extreme points which are not absolute.

**Lemma 3.8.** *[8, Lemma 2.3] Let $\mathcal{S}$ be an operator system and let $\varphi \in \mathrm{UCP}(\mathcal{S}, B(\mathcal{H}))$ be a pure UCP map which is not maximal. Then, given $s_0 \in \mathcal{S}, h \in \mathcal{H}$ such that $\varphi$ is not maximal at $(s_0, h)$, there exists a pure $\psi \in \mathrm{UCP}(\mathcal{S}, B(\mathcal{H} \oplus \mathbb{C}))$ which is maximal at $(s_0, h)$ and such that $\varphi(s) = P_{\mathcal{H}}\, \psi(s)|_{\mathcal{H}}$.*

We can now give the following useful description for absolute matrix extreme points and matrix extreme points in $\mathrm{UCP}(\mathcal{S})$:

- Absolute matrix extreme points are precisely the restrictions of finite dimensional boundary representations, and they cannot be written as a compression of another matrix extreme point from higher levels (because they are maximal).

- Matrix extreme points in $\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$ which are not absolute, are pure and not maximal UCP maps which are compressions of UCP maps in $\mathrm{UCP}(\mathcal{S}, \mathbb{M}_{n+1})$. Moreover, by [8, Theorem 2.4], they are always of the form $V^* \pi(\cdot) V$ for a (not necessarily finite dimensional) boundary representation).

**Matrix extremality in general compact matrix convex sets** - Let $K = (K_n)_{n=1}^{\infty}$ be a compact matrix convex set. By [22, Proposition 3.5], it is matrix affine homeomorphic to UCP $(A(K))$. Because matrix affine homeomorphisms preserve matrix extreme and absolute matrix extreme points, we can give the following description for matrix extreme points without considering $A(K)$ directly.

**Proposition 3.9.** *Let $K = (K_n)_{n=1}^{\infty}$ be a compact matrix convex set in a locally convex vector space $V$. A matrix extreme point $v \in \partial K_n$ satisfies exactly one of the following:*

- *$v$ is an absolute matrix extreme point.*
- *There exists some $u \in \partial K_{n+1}$ such that $v = \gamma^* u \gamma$ for some $\gamma \in \mathbb{M}_{n,n+1}$.*

Namely, matrix extreme point is either absolute, or it dilates to another matrix extreme point at the next level.

### 3.1.1 $N$-minimal matrix convex sets

For $N$-generated matrix convex sets of matrices, we have the following useful lemma due to Kriel:

**Lemma 3.10.** *[14, Lemma 1.14] Let $v$ be a $d$-tuple of $m \times m$ self-adjoint matrices and let $T$ be a subset of $\left(\mathbb{M}_n^d\right)_{n=1}^{\infty}$ which consists of self-adjoint matrices. If $v \in \mathrm{co}(T)$, then there exists $v_1, \ldots, v_{2dm^2+1} \in T$ and matrices $\gamma_1, \ldots, \gamma_{2dm^2+1}$ of appropriate sizes, such that $I_m = \sum_{j=1}^{2dm^2+1} \gamma_j^* \gamma_j$ and $v = \sum_{j=1}^{2dm^2+1} \gamma_j^* v_j \gamma_j$.*

Namely, for $d$-tuples of $m \times m$ matrices, any matrix convex combination can be reduced to $2dm^2 + 1$ summands or less.

The fact the matrices were self-adjoint was used to show that $N > 2dm^2 + 1$ self-adjoint tuples of matrices of sizes $m^2 \times m^2$ (with an additional assumption that they are direct sums of two $m \times m$ matrices) must be $\mathbb{R}$-linearly independent. In order to fit the case of general matrices, one may replace that number by $4dm^2 + 1$ and get similar results. The important thing is that we can bound the number of summand in each such matrix convex combination, and use this to we can deduce the following:

**Proposition 3.11.** *Let $K = (K_n)_{n=1}^{\infty}$ be a compact matrix convex set in $\mathbb{M}^d$. Then $\mathrm{co}(K_N)$ is compact for all $N \in \mathbb{N}$.*

*Proof.* Set $L = \mathrm{co}(K_N)$, and let $v \in \mathbb{M}_n^d$ be a limit point of $L_n$. If $n = N$, we have that since $L_N = K_N$, and $K_N$ is compact, then $v \in K_N$. Otherwise, we know that $v$ is a limit of matrix convex combinations from $K_N$ which consists of a bounded number of summands. Therefore, each summand admits a limit point in $K_N$ because $K_N$ is compact, which proves that $v$ is a matrix convex combination of points in $K_N$, so $v \in L_n$. □

**Corollary 3.12.** *Let $K = (K_n)_{n=1}^{\infty}$ be a compact matrix convex set. Then $\mathcal{W}^{N\text{-}min}(K)$ is compact for all $N \in \mathbb{N}$.*

*Proof.* Let $L = (L_n)_{n=1}^{\infty}$ be matrix convex such that $K_N \subset L_N$. By matrix convexity, $\mathrm{co}(K_N) \subset L$. This means that $\mathrm{co}(K_N) = \mathcal{W}^{N\text{-min}}(K)$ by definition. Therefore, the set is compact by Proposition 3.11. □

By the Webster-Winkler Krein-Milman theorem, we know that $K$ can be recovered from $\partial K$ by taking limits of matrix convex combinations. One should hope that $N$-generated compact matrix convex sets of matrices would be minimal, since all of the information should lie at level $N$. This is indeed the case:

**Proposition 3.13.** *Let $K = (K_n)_{n=1}^{\infty}$ be a compact matrix convex set in $\mathbb{M}^d$. Then $K = \mathcal{W}^{N\text{-}min}(K)$ if and only if it is $n$-generated for some $n \leq N$.*

*Proof.* Assume that $K = \mathcal{W}^{N\text{-}\min}(K) = \text{co}(K_N)$. Therefore, any point $v \in K_n$ for $n > N$ is a proper matrix convex combination of points from levels $N$ or below, so it is not matrix extreme, and that proves that $K$ is $n$-generated for some $n \leq N$. Conversely, assume that $K$ is $n$-generated for $n \leq N$, and let $L$ be a matrix convex set such that $L_N = K_N$. If $v \in K_n$, then for some $u \in K_{N-n}$ we have that $v \oplus u \in K_N = L_N$, which means that $v \in L_n$ is a matrix convex combination of $v \oplus u$ (which is just the compression). This means that $K_n \subset L_n$ and by convexity:

$$\text{co}(K_n) \subset L.$$

Lastly, we claim that $K = \text{co}(K_n)$, with similar arguments as in Proposition 3.11. If $v \in \partial K_m$ for $m < n$, then once again $v$ is a matrix convex combination of points from $K_n$, so any limit point of matrix convex combinations from $\partial K$ can be written as a matrix convex combination with a constant amount of summands, all from $K_n$. So $K_n$ being compact implies that any limit of such points is already in $\text{co}(K_n)$, and therefore:

$$K = \overline{\text{co}}(\partial K) = \overline{\text{co}}(K_n) = \text{co}(K_n) \subset L$$

which proves that $K$ is $N$-minimal. □

## 3.2 $N$-generated matrix state spaces

Equipped with the better understanding of matrix extreme points in UCP and the importance of $N$-generated sets, we prove several results for operator systems and their associated matrix state spaces. Recall that due to Webster and Winkler, any operator system $\mathcal{S}$ is completely order isomorphic to $A(\text{UCP}(\mathcal{S}))$. For an operator system $\mathcal{R}$, it is only natural that an isomorphism (namely, matrix affine homeomorphism) of $\text{UCP}(\mathcal{S})$ and $\text{UCP}(\mathcal{R})$ should give us a complete order isomorphism of the operator systems. This is proved below.

**Proposition 3.14.** *Let $\mathcal{S}, \mathcal{R}$ be two operator systems, and assume that $T = (T_n)_{n=1}^{\infty}$ is a matrix affine homeomorphism of $\text{UCP}(\mathcal{S}), \text{UCP}(\mathcal{R})$. Then, $T$ induces a complete order isomorphism of $\mathcal{S}$ and $\mathcal{R}$.*

*Proof.* Define a mapping $T_* : A(\text{UCP}(\mathcal{R})) \to A(\text{UCP}(\mathcal{S}))$ defined by $T_*F = (F \circ T_n)_{n=1}^{\infty}$. We claim that it is a complete order isomorphism of the two operator systems.

In order to do so we use the identification of $\mathbb{M}_n(A(\text{UCP}(\mathcal{R})))$ with $A(\text{UCP}(\mathcal{R}), \mathbb{M}_n)$, and similarly we do for $\mathcal{S}$. In that case, $(T_*)_n$ takes the form of the mapping:

$$(T_*)_n F = (F_m \circ T_m)_{m=1}^{\infty}$$

for all $F \in A(\mathrm{UCP}(\mathcal{R}), \mathbb{M}_n)$. By definition, $F$ is positive if and only if $F_m(\phi)$ is positive for all $\phi \in \mathrm{UCP}(\mathcal{R}, \mathbb{M}_m)$, and since $T$ is a bijection, $F_m$ is positive if and only if $F_m \circ T_m(\psi)$ is positive for all $\psi \in \mathrm{UCP}(\mathcal{S}, \mathbb{M}_m)$ which means that $(T_*)_n F$ is positive. $T_*$ being unital (the unit does not depend on the argument put into it) means that this map is UCP.

Finally, using the same arguments for $T^{-1}$ shows that the map is a complete order isomorphism. Finally, $\delta_{\mathcal{S}}^{-1} \circ T_* \circ \delta_{\mathcal{R}}$ is a complete order isomorphism of $\mathcal{S}$ and $\mathcal{R}$ as a composition of complete order isomorphisms. □

Next, we move to discuss what is the matrix convex set equivalent of an $N$-order isomorphism.

Let $\mathcal{S}, \mathcal{R}$ be operator systems. The collection $(L(\mathcal{S}, \mathbb{M}_n))_{n=1}^{\infty}$ of linear maps from $\mathcal{S}$ to $\mathbb{M}_n$ for all $n \in \mathbb{N}$, is matrix convex. Indeed, a matrix convex combination of linear maps is a linear map. Given a linear map $\phi : \mathcal{S} \to \mathcal{R}$, we can consider the pullback maps $\boldsymbol{\phi}^* = (\phi_n^*)_{n=1}^{\infty}$ such that $\phi_n^* : L(\mathcal{R}, \mathbb{M}_n) \to L(\mathcal{S}, \mathbb{M}_n)$ given by $\phi_n^* \psi = \psi \circ \phi$.

**Lemma 3.15.** *The collection $\boldsymbol{\phi}^*$ is a matrix affine map of $(L(\mathcal{R}, \mathbb{M}_n))_{n=1}^{\infty}$ into $(L(\mathcal{S}, \mathbb{M}_n))_{n=1}^{\infty}$.*

*Proof.* By direct calculation, if $\psi = \sum_i \gamma_i^* \psi_i \gamma_i \in L(\mathcal{R}, \mathbb{M}_n)$, then:

$$\phi^* \psi = \psi \circ \phi = \sum_i \gamma_i^* \psi_i \circ \phi \gamma_i = \sum_i \gamma_i^* \phi_{n_i}^* \psi_i \gamma_i$$

□

In general, if $\psi \in \mathrm{UCP}(\mathcal{R}, \mathbb{M}_n)$ and $\phi : \mathcal{S} \to \mathcal{R}$ is linear, $\phi^* \psi$ might not be UCP. In order to achieve this, we have to put more assumptions on $\phi$.

**Proposition 3.16.** *Let $\phi : \mathcal{S} \to \mathcal{R}$ be an $N$-order isomorphism, and let $k \leq N$. Then, $\phi_k^*$ restricts to a bijection of $\partial\mathrm{UCP}(\mathcal{R}, \mathbb{M}_k)$ and $\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_k)$.*

*Proof.* We first note that for $k \leq N$, if $\psi \in \mathrm{UCP}(\mathcal{R}, \mathbb{M}_k)$, the composition $\phi^* \psi = \phi \circ \psi$ is an $N$-positive map from $\mathcal{S}$ to $\mathbb{M}_k$ (so in particular, it is $k$-positive by Lemma 2.5). Since a $k$-positive map to $\mathbb{M}_k$ is completely positive (see Theorem 2.7), we get that $\phi^* \psi \in \mathrm{UCP}(\mathcal{S}, \mathbb{M}_k)$.

Now we move to show that if $\psi$ is matrix extreme in some level below $N$, then $\phi^* \psi$ is a matrix extreme point. Assume that $\phi^* \psi$ is a proper matrix convex combination of the form:

$$\phi^* \psi(\cdot) = \sum_i \gamma_i^* \psi_i(\cdot) \gamma_i$$

for $\psi_i \in \mathrm{UCP}(\mathcal{S}, \mathbb{M}_{k_i})$, $\gamma_i \in \mathbb{M}_{k_i, k}$ for $i = 1, \ldots, \ell$ satisfying $\sum_i \gamma_i^* \gamma_i = \mathbb{1}_k$, and that $k_i \leq k$ for all $i$. Since $\phi^{-1}$ is also $N$ positive, we have that:

$$\psi(\cdot) = \left(\phi^{-1}\right)^* \phi^* \psi(\cdot) = \sum_i \gamma_i^* \left(\phi^{-1}\right)^* \psi_i(\cdot) \gamma_i.$$

Note, that since $(\phi^{-1})^* \psi_i \in \mathrm{UCP}(\mathcal{R}, \mathbb{M}_{k_i})$, the right hand side is a proper matrix convex combination in $\mathrm{UCP}(\mathcal{R})$. Since $\psi$ is an extreme point, we get that $n_i = i$ and that $\psi = u_i^* (\phi^{-1})^* \psi_i u_i$ for some unitary $u_i$. But this means that $\phi^* \psi = u_i^* \psi_i u_i$, which proves that it is an extreme point.

Lastly, $\phi^* : \partial\mathrm{UCP}(\mathcal{R}, \mathbb{M}_k) \to \partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_k)$ is a bijection because $(\phi^{-1})^*$ is easily seen to be its inverse by using the same arguments. □

The arguments in the proof can be used to show that the pullback map gives a bijection of $\mathrm{UCP}(\mathcal{R}, \mathbb{M}_k)$ and $\mathrm{UCP}(\mathcal{S}, \mathbb{M}_k)$ for all $k \leq N$. In the next proposition, we extend this result for levels higher than $N$.

**Proposition 3.17.** *Let $\phi : \mathcal{S} \to \mathcal{R}$ be an $N$-order isomorphism. Then, $\boldsymbol{\phi}^*$ restricts to a matrix affine homeomorphism of $\overline{\mathrm{co}}\{\partial\mathrm{UCP}(\mathcal{R}, \mathbb{M}_k)\}_{k=1}^N$ and $\overline{\mathrm{co}}\{\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_k)\}_{k=1}^N$.*

*Proof.* By Lemma 3.15 and Proposition 3.16, we have that for $\psi \in \mathrm{co}(\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_k))_{k=1}^N$, the map $\phi_n^*\psi$ is a matrix convex combination matrix extreme points of $\mathrm{UCP}(\mathcal{S})$, which are from level $N$ or below. Therefore, it is UCP and in $\mathrm{co}(\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_k))_{k=1}^N$.

Before considering the closure of that set, note that for all $n \in \mathbb{N}$, we have that $\phi_n^* : L(\mathcal{R}, \mathbb{M}_n) \to L(\mathcal{S}, \mathbb{M}_n)$ is continuous with respect to the weak* topology. Indeed, $\psi_m$ converges to $\psi$ in $L(\mathcal{R}, \mathbb{M}_n)$ if and only if $\psi_m(r)$ converges to $\psi(r)$ for all $r \in \mathcal{R}$. Therefore, if $\psi_m$ converges to $\psi$ we have that $\phi_n^*\psi_m(s) = \psi_m(\phi(s))$ converges to $\psi(\phi(s)) = \phi_n^*\psi(s)$ for all $s \in \mathcal{S}$, and thus $\phi_n^*\psi_m$ converges to $\phi_n^*\psi$ in $L(\mathcal{S}, \mathbb{M}_n)$.

From this we can deduce that if $\psi \in \overline{\mathrm{co}}(\partial\mathrm{UCP}(\mathcal{R}, \mathbb{M}_k))_{k=1}^N$ is a mapping into $\mathbb{M}_n$, then $\psi$ is a limit of a net $(\psi_m)_m$ in $\mathrm{co}(\partial\mathrm{UCP}(\mathcal{R}, \mathbb{M}_k))_{k=1}^N$. Therefore:

$$\phi_n^*\psi = \phi_n^*(\lim_m \psi_m) = \lim_m \phi_n^*\psi_m,$$

from which it follows that $\phi_n^*\psi \in \overline{\mathrm{co}}(\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_k))_{k=1}^N$. This proves that the restriction of $\boldsymbol{\phi}^*$ to $\overline{\mathrm{co}}(\partial\mathrm{UCP}(\mathcal{R}, \mathbb{M}_k))_{k=1}^N$ is matrix affine and continuous into $\overline{\mathrm{co}}(\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_k))_{k=1}^N$. Similar arguments for $(\boldsymbol{\phi}^{-1})^*$ shows it is a matrix affine homeomorphism. □

For $N$-generated matrix state spaces, we can deduce the following:

**Theorem 3.18.** *Let $\mathcal{S}, \mathcal{R}$ be operator systems such that $\mathrm{UCP}(\mathcal{S}), \mathrm{UCP}(\mathcal{R})$ are $N_{\mathcal{S}}$-generated and $N_{\mathcal{R}}$-generated for some $N_{\mathcal{S}}, N_{\mathcal{R}} \in \mathbb{N}$. Let $N = \max\{N_{\mathcal{S}}, N_{\mathcal{R}}\}$. Then $\mathcal{S}$ and $\mathcal{R}$ are $N$-order isomorphic if and only if they are completely order isomorphic.*

*Proof.* By Proposition 3.17, we have that $\boldsymbol{\phi}^*$ restricts to a matrix affine homeomorphism of $\overline{\mathrm{co}}(\partial\mathrm{UCP}(\mathcal{R}, \mathbb{M}_k))_{k=1}^N$ and $\overline{\mathrm{co}}(\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_k))_{k=1}^N$. Because the sets are $N_{\mathcal{R}}$ and $N_{\mathcal{S}}$ generated, we use the Webster-Winkler Kreim-Milman Theorem (see 2.19), and conclude that the restriction is a matrix affine homeomorphism of $\mathrm{UCP}(\mathcal{R})$ and $\mathrm{UCP}(\mathcal{S})$. By Proposition 3.14, we get that $\delta_{\mathcal{R}}^{-1} \circ (\boldsymbol{\phi}^*)_* \circ \delta_{\mathcal{S}}$ is a complete order isomorphism of $\mathcal{S}$ and $\mathcal{R}$. But note that for all $s \in \mathcal{S}$ and $\psi \in \mathrm{UCP}(\mathcal{R}, \mathbb{M}_n)$:

$$((\boldsymbol{\phi}^*)_* \circ \delta_{\mathcal{S}}(s))_n(\psi) = (\delta_{\mathcal{S}}(s))_n\phi_n^*(\psi) = \psi(\phi(s)) = (\delta_{\mathcal{R}}(\phi(s)))_n(\psi)$$

which means that $\phi = \delta_{\mathcal{R}}^{-1} \circ (\boldsymbol{\phi}^*)_* \circ \delta_{\mathcal{S}}$. Therefore, $\phi$ is a complete order isomorphism. □

We can even show a variation of this theorem for unital and $N$-positive maps which do not have an $N$-positive inverse.

**Theorem 3.19.** *Let $\mathcal{S}$ be an operator system such that $\mathrm{UCP}(\mathcal{S})$ is $N$-generated. Then for any operator system $\mathcal{R}$, if $\phi : \mathcal{R} \to \mathcal{S}$ is unital and $N$-positive, it is UCP.*

*Proof.* By considering the pullback map, we show that $\boldsymbol{\phi}^*$ restricts to a continuous matrix affine map $\boldsymbol{\phi}^* : \mathrm{UCP}(\mathcal{S}) \to \mathrm{UCP}(\mathcal{R})$. In order to do so, recall that $\boldsymbol{\phi}^*$ is matrix affine and continuous, so we just need to show that for all $\psi \in \mathrm{UCP}(\mathcal{S})$, $\phi^*\psi \in$

UCP $(\mathcal{R})$. Indeed, because UCP $(\mathcal{S})$ is $N$-generated, we get by the Webster-Winkler Krein-Milman theorem, that it is a limit of matrix convex combinations consisting of matrix extreme points, which all lie at levels below $N$. because $\boldsymbol{\phi}^*$ is matrix extreme and continuous, we get that $\phi^*\psi$ is a limit point of matrix convex combinations of points in UCP $(\mathcal{R})$. Since UCP $(\mathcal{R})$ is closed, we have that $\phi^*\psi$ is UCP.

Now, set $(\boldsymbol{\phi}^*)_* : A(\text{UCP}(\mathcal{R})) \to A(\text{UCP}(\mathcal{S}))$ by $(\boldsymbol{\phi}^*)_* F := (F_n \circ \phi_n^*)_{n=1}^{\infty}$. The same calculation as in the proof of Proposition 3.14 shows that this map is a completely positive map of the two operator systems.

Lastly, we have that $\phi = \delta_{\mathcal{S}}^{-1} \circ (\boldsymbol{\phi}^*)_* \circ \delta_{\mathcal{R}}$, which is a composition of completely positive maps, which makes $\phi$ UCP. □

It's worth noting that this is a generalization of Theorem 2.7 for operator systems with finitely generated matrix state spaces. It shows that if an operator system has finite dimensional boundary representations of bounded dimensions, then maps into it behave like maps into some $\mathbb{M}_n$ for a sufficiently large $n$.

## 3.3 $N$-generated matrix ranges

For the case where $\mathcal{S}, \mathcal{R}$ are generated by $d$-tuples, we consider their matrix ranges, and apply the results we have for general operator systems.

**Proposition 3.20.** *Let $\mathcal{S} = \mathcal{S}(\mathbf{s}), \mathcal{R} = \mathcal{S}(\mathbf{r})$ be two operator systems generated by $d$-tuples, and let $\phi : \mathcal{S} \to \mathcal{R}$ be an $N$-order isomorphism such that $\phi(s_i) = r_i$. Then:*

$$\mathcal{W}_n(\mathbf{s}) = \mathcal{W}_n(\mathbf{r})$$

*for all $n \leq N$.*

*Proof.* For all $n \leq N$, $\phi_n^* : \text{UCP}(\mathcal{R}, \mathbb{M}_n) \to \text{UCP}(\mathcal{S}, \mathbb{M}_n)$ is a bijection. Therefore:

$$\mathcal{W}_n(\mathbf{r}) = \{(\psi(r_1), \ldots, \psi(r_d)) | \psi \in \text{UCP}(\mathcal{R}, \mathbb{M}_n)\} = \{(\psi \circ \phi(s_1), \ldots, \psi \circ \phi(s_d)) | \psi \in \text{UCP}(\mathcal{R}, \mathbb{M}_n)\}$$

$$= \{(\varphi(s_1), \ldots, \varphi(s_d)) | \varphi \in \text{UCP}(\mathcal{S}, \mathbb{M}_n)\} = \mathcal{W}_n(\mathbf{s})$$

□

Note that in Example 2.22, we proved that for $\mathcal{S} = \mathcal{S}(\mathbf{s})$, UCP $(\mathcal{S})$ and $\mathcal{W}(\mathbf{s})$ are matrix affine homeomorphic. Therefore, UCP $(\mathcal{S})$ is $N$-generated if and only if $\mathcal{W}(\mathbf{s})$ is $N$-generated. This allows us to shift the two perspectives when one is more convenient than the other.

The next example is of two $d$-tuples such that $\mathcal{W}(\mathbf{s})$ is $N_{\mathcal{S}}$-generated and $\mathcal{W}(\mathbf{r})$ is $N_{\mathcal{R}}$-generated, but the corresponding operator systems are $(\max\{N_{\mathcal{S}}, N_{\mathcal{R}}\} - 1)$-order isomorphic even though they are not completely order isomorphic (thereby proving that in general, taking the maximum cannot be improved).

Before presenting it, we consider the following lemma.

**Lemma 3.21.** *Let $\mathbf{s}, \mathbf{r}$ be two $d$-tuples. Then, $\mathcal{W}_1(\mathbf{s}) = \mathcal{W}_1(\mathbf{r})$ if and only if the tuples are 1-order equivalent.*

*Proof.* By [7, Theorem 5.1], if the tuples are 1-order equivalent then $\mathcal{W}_1(\mathbf{s}) = \mathcal{W}_1(\mathbf{r})$, so we only need to show the converse. We assume the equality of the first level in the matrix range, and in order to show that the mapping $s_i \mapsto r_i$ extends to a well defined 1-order isomorphism of the operator systems, we use the fact that positive maps are self-adjoint, so such a mapping must also satisfy that $s_i^*$ will be mapped to $r_i^*$. Therefore, we define $\phi : \mathcal{S}(\mathbf{s}) \to \mathcal{S}(\mathbf{r})$ by $\phi(s_i) = r_i, \phi(s_i^*) = r_i^*$ and $\phi(\mathbb{1}_{\mathbf{s}}) = \mathbb{1}_{\mathbf{r}}$, and prove that its linear extension is positive.

Let $a_0\mathbb{1}_{\mathcal{S}} + \sum_i a_i s_i + b_i s_i^*$ be positive in $\mathcal{S}$. In order to prove that:

$$\phi\left(a_0\mathbb{1}_{\mathcal{S}} + \sum_i a_i s_i + b_i s_i^*\right) = a_0\mathbb{1}_{\mathcal{R}} + \sum_i a_i r_i + b_i r_i^*$$

is positive in $\mathcal{R}$, it is sufficient to show that for all states $\psi : \mathcal{R} \to \mathbb{C}$:

$$\psi\left(a_0\mathbb{1}_{\mathcal{R}} + \sum_i a_i r_i + b_i r_i^*\right) = a_0 + \sum_i a_i\psi(r_i) + b_i\psi(r_i)^* \geq 0$$

But note that since $\psi : \mathcal{R} \to \mathbb{C} = \mathbb{M}_1$ is positive, it is completely positive, so:

$$(\psi(r_1), \ldots, \psi(r_d)) \in \mathcal{W}_1(\mathbf{r}) = \mathcal{W}_1(\mathbf{s}).$$

Therefore, $(\psi(r_i), \ldots, \psi(r_d)) = (\rho(s_1), \ldots, \rho(s_d))$ for a state $\rho : \mathcal{S} \to \mathbb{C}$. So:

$$a_0 + \sum_i a_i\psi(r_i) + b_i\psi(r_i)^* = a_0 + \sum_i a_i\rho(r_i) + b_i\rho(s_i)^* = \rho\left(a_0 + \sum_i a_i s_i + b_i s_i^*\right) \geq 0$$

because the argument is positive in $\mathcal{S}$. Therefore $\phi$ is a positive map (which also means it is well defined on the operator system), and $\phi^{-1}$ will be positive as well by using the same argument. □

**Example 3.22.** Let $\mathcal{H} = L^2(\bar{\mathbb{B}}_2)$ where $\bar{\mathbb{B}}_2$ is the closed Euclidean unit ball in $\mathbb{R}^2$. Define $M_{x_1}, M_{x_2} \in B(\mathcal{H})$ by $M_{x_i} f(x_1, x_2) = x_i f(x_1, x_2)$. Then we have that $M_{x_1}, M_{x_2}$ are a pair of self adjoint and commuting operators. By [7, Corollary 4.4], the matrix range of the pair is a minimal one. Namely:

$$\mathcal{W}(M_{x_1}, M_{x_2}) = \mathcal{W}^{\min}(\sigma(M_{x_1}, M_{x_2})).$$

Moreover, [7, Theorem 2.7] tells us that $\mathcal{W}_1(M_{x_1}, M_{x_2}) = \sigma(M_{x_1}, M_{x_2}) = \bar{\mathbb{B}}_2$. On the other hand, consider the tuple $(F_1, F_2) \in B(\mathbb{C}^2)^2$ of the matrices (see Chapter 5 in [19] for a wider context):

$$F_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad F_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

It is easily verified that this is a tuple of anti-commuting unitaries. Applying [19, Corollary 5.9] we get that the mapping $M_{x_i} \mapsto F_i$ extends to a unital and positive map from $\mathcal{S}(M_{x_1}, M_{x_2})$ to $\mathcal{S}(F_1, F_2)$, and therefore:

$$\mathcal{W}_1(F_1, F_2) \subset \mathcal{W}_1(M_{x_1}, M_{x_2}).$$

On the other hand, whenever $\xi = (\xi_1, \xi_2) \in \mathbb{C}^2$ with $|\xi_1|^2 + |\xi_2|^2 = 1$, we get that:

$$(\xi^* F_1 \xi, \xi^* F_2 \xi) \in \mathcal{W}_1(F_1, F_2).$$

For a given $\theta \in [0, 2\pi]$, set $\xi = \left(r_1 e^{i\theta_1}, r_2 e^{i\theta_2}\right)$ such that:

$$r_1 = \sqrt{\frac{1 + \cos(\theta)}{2}}, \quad r_2 = \sqrt{\frac{1 - \cos(\theta)}{2}}, \quad \cos(\theta_1 - \theta_2) = \operatorname{sgn}(\sin(\theta)).$$

By a direct calculation we have that:

$$\|\xi\|^2 = 1, \quad (\xi^* F_1 \xi, \xi^* F_2 \xi) = (\cos(\theta), \sin(\theta))$$

which means that the unit circle is contained in $\mathcal{W}_1(F_1, F_2)$. This set being convex means that:

$$\mathcal{W}_1(M_{x_1}, M_{x_2}) = \bar{\mathbb{B}}_2 \subset \mathcal{W}_1(F_1, F_2).$$

Since we have already proved the other inclusion, we get equality.

By Lemma 3.21, we get that the tuples are 1-order equivalent, which means that the operator systems generated are 1-order isomorphic.

Note, that because $\mathcal{W}(M_{x_1}, M_{x_2})$ is the minimal matrix convex set which has the ball at the first level, it must be 1-generated, since we can write it as:

$$\mathcal{W}(M_{x_1}, M_{x_2}) = \operatorname{co}(\mathcal{W}_1(M_{x_1}, M_{x_2}))$$

which means that any point at level 2 or higher is a proper matrix convex combination of points from level 1, so it is not extreme.

In Theorem 5.6, we show that $\mathcal{W}(F_1, F_2)$ is 2-generated. From this it follows that the operator systems generated cannot be 2-order isomorphic, or otherwise $\partial\mathcal{W}_2(M_{x_1}, M_{x_2})$ would be non-empty, by Proposition 3.16.

**Proposition 3.23.** *Let* $\mathbf{s}$ *be a tuple such that* $\mathcal{W}(\mathbf{s})$ *is* $N$*-generated. Then:*

$$\mathcal{W}(\mathbf{s}) = \mathcal{W}^{N\text{-}min}(\mathbf{s})$$

*Proof.* $\mathcal{W}(\mathbf{s})$ is an $N$-generated compact matrix convex set in $\mathbb{M}^d$. The equality is then due to Proposition 3.13. □

We can now use the last results to give the following for tuples.

**Theorem 3.24.** *Let* $\mathbf{s}, \mathbf{r}$ *be two* $d$*-tuples such that* $\mathcal{W}(\mathbf{s}), \mathcal{W}(\mathbf{r})$ *are* $N_{\mathbf{s}}$*-generated and* $N_{\mathbf{r}}$*-generated, and set* $N = \max\{N_{\mathbf{s}}, N_{\mathbf{r}}\}$*. The following are equivalent:*

1. $\mathbf{s}$ *and* $\mathbf{r}$ *are* $N$*-order equivalent.*
2. $\mathbf{s}$ *and* $\mathbf{r}$ *are completely order equivalent.*
3. $\mathcal{W}_N(\mathbf{s}) = \mathcal{W}_N(\mathbf{r})$.
4. $\mathcal{W}(\mathbf{s}) = \mathcal{W}(\mathbf{r})$.

*Proof.* (1) $\iff$ (2) follows from Theorem 3.18, and (1) $\implies$ (3) follows from [7, Theorem 5.1]. We now prove (3) $\implies$ (4). Note that for both matrix ranges we have:

$$\mathcal{W}(\mathbf{s}) = \mathcal{W}^{N_{\mathbf{s}}\text{-min}}(\mathbf{s}), \quad \mathcal{W}(\mathbf{r}) = \mathcal{W}^{N_{\mathbf{r}}\text{-min}}(\mathbf{r}),$$

and thus, equality of $\mathcal{W}_N(\mathbf{s}) = \mathcal{W}_N(\mathbf{r})$ means in that for and $v \in \mathcal{W}_n(\mathbf{s})$ for $n \leq N_{\mathbf{s}}$, it is a compression of $v \oplus u \in \mathcal{W}_N(\mathbf{s}) = \mathcal{W}_N(\mathbf{r})$ for some $u \in \mathcal{W}_{N-n}(\mathbf{s})$. Therefore, $v$ is a compression of a point in $\mathcal{W}_N(\mathbf{r})$ and thus in $\mathcal{W}_n(\mathbf{r})$. The same argument works the other direction and thus $\mathcal{W}_n(\mathbf{s}) = \mathcal{W}_n(\mathbf{r})$ for all $n \leq N$. In particular, $\mathcal{W}_{N_{\mathbf{s}}}(\mathbf{s}) = \mathcal{W}_{N_{\mathbf{s}}}(\mathbf{r})$, and by minimality we get:

$$\mathcal{W}(\mathbf{s}) \subset \mathcal{W}(\mathbf{r})$$

and again, by using the same arguments for $\mathbf{r}$, we get the second inclusion, and thus equality. Lastly, (4) $\iff$ (2) follows also from [7, Theorem 5.1]. $\square$

# Chapter 4

# Operator Systems Generated by Unitaries

## 4.1 Order Equivalence and $*$-Isomorphisms

In this chapter $\mathcal{S} = \mathcal{S}(\mathbf{s}), \mathcal{R} = \mathcal{S}(\mathbf{r})$ are two operator systems generated by tuples of unitaries. Unlike general operator systems (even those which are generated by some tuple), unitaries have some "rigidity" properties which we want to use in light of the results from the previous chapter. First, we show that for unitary tuples, complete order equivalence means $*$-isomorphic and vice versa.

Recall that for an operator system $\mathcal{S}$, and a $C^*$-algebra $\mathcal{B}$, we say that a map $\phi \in \mathrm{UCP}(\mathcal{S}, \mathcal{B})$ has the *unique extension property* if $\phi$ has a unique extension to a map $\Phi \in \mathrm{UCP}(C^*(\mathcal{S}), \mathcal{B})$, and this map is a $*$-homomorphism.

**Theorem 4.1.** *Let $\mathbf{s}, \mathbf{r}$ be tuples of unitaries, and assume that $s_i \mapsto r_i$ defines a UCP of $\mathcal{S} = \mathcal{S}(\mathbf{s})$ to $\mathcal{R} = \mathcal{S}(\mathbf{r})$. Then, this map has the unique extension property.*

*Proof.* Denote the mapping by $\phi$. We may assume that $C^*(\mathbf{s}), C^*(\mathbf{r})$ are concrete subalgebras of operators on Hilbert spaces $\mathcal{H}, \mathcal{K}$ respectively. In that case, we may apply Arveson's extension theorem (see [2, Corollary 1.2.3]) and find some UCP $\Phi : C^*(\mathbf{s}) \to C^*(\mathbf{r})$ which extends $\phi$. Then, we apply Stinespring's dilation theorem (see [21, Theorem 1]) to find a Hilbert space $\mathcal{K}'$, a $*$-homomorphism $\pi : C^*(\mathbf{s}) \to B(\mathcal{K}')$ and an isometry $V : \mathcal{K} \to \mathcal{K}'$ such that for all $s \in C^*(\mathbf{s})$:

$$\Phi(s) = V^* \pi(s) V.$$

Since $V$ is an isometry we may also identify $\mathcal{K}$ with the closed subspace $V\mathcal{K} \subset \mathcal{K}'$. This means that by decomposing $\mathcal{K}' = \mathcal{K} \oplus \mathcal{K}^\perp$, we can write for all $a \in C^*(\mathbf{s})$:

$$\pi(a) = \begin{pmatrix} \Phi(a) & \beta \\ \gamma & \delta \end{pmatrix}.$$

In particular, for all $i$, we have that $s_i$ and $\phi(s_i)$ are both unitaries, which means that:

$$\pi(s_i)^*\pi(s_i) = \begin{pmatrix} \phi(s_i)^*\phi(s_i) + \gamma_i^*\gamma_i & \phi(s_i)^*\beta_i + \gamma_i^*\delta_i \\ \beta_i^*\phi(s_i) + \delta_i^*\gamma_i & \beta_i^*\beta_i + \delta_i^*\delta_i \end{pmatrix} = \begin{pmatrix} \mathrm{Id}_{\mathcal{K}} + \gamma_i^*\gamma_i & \phi(s_i)^*\beta_i + \gamma_i^*\delta_i \\ \beta_i^*\phi(s_i) + \delta_i^*\gamma_i & \beta_i^*\beta_i + \delta_i^*\delta_i \end{pmatrix}$$

$$= \begin{pmatrix} \mathrm{Id}_{\mathcal{K}} & 0 \\ 0 & \mathrm{Id}_{\mathcal{K}^\perp} \end{pmatrix}$$

and thus we deduce that $\gamma_i^*\gamma_i = 0$ so $\gamma_i = 0$. Applying the same calculation for $\pi(s_i)\pi(s_i)^*$ yields $\beta_i = 0$, and thus:

$$\pi(s_i) = \begin{pmatrix} r_i & 0 \\ 0 & \delta_i \end{pmatrix}$$

for all $i = 1, \ldots, d$. From this we can easily see that $\Phi$ is a $*$-homomorphism from $C^*(\mathbf{s})$ to $C^*(\mathbf{r})$. □

When a mapping such as in the theorem has a UCP inverse, we have the following theorem.

**Theorem 4.2.** *The mapping $s_i \mapsto r_i$ extends to a complete order isomorphism of $\mathcal{S}$ and $\mathcal{R}$ if and only if it extends to a $*$-isomorphism of $C^*(\mathbf{s})$ and $C^*(\mathbf{r})$. In particular, the $*$-isomorphism is the unique UCP extension of that mapping.*

*Proof.* A $*$-isomorphism is already a complete order isomorphism, and it restricts to one between the operator systems. Therefore, we will only prove the converse. By our assumptions, there exits a UCP with a UCP inverse, which maps $\mathbf{s}$ to $\mathbf{r}$. By Theorem 4.1, both mappings have the unique extension property, so they both extend to a $*$-homomorphism of the generated $C^*$-algebras. Those extensions are completely determined by $\mathbf{s}$ and $\mathbf{r}$, which proves that they are inverses of each other. This completes the proof. □

This shows us that equivalence of the complete order structure generated by unitary tuples encloses all of the information about the algebras that they generate.

## 4.2 Tuple Equivalence Using Dilation Theory

The existence of UCP maps which map an operator system to the other is deeply connected to dilation problems.

**Definition 4.3.** Let $\mathbf{s}, \mathbf{r}$ be two $d$-tuples generating $C^*(\mathbf{s}), C^*(\mathbf{r})$ respectively. We say that $\mathbf{s}$ *dilates* to $\mathbf{r}$ if there are $*$-isomorphisms $\pi : C^*(\mathbf{s}) \to \mathcal{A} \subset B(\mathcal{H}), \rho : C^*(\mathbf{r}) \to \mathcal{B} \subset B(\mathcal{K})$, and an isometry $V : \mathcal{H} \to \mathcal{K}$ such that:

$$\pi(s_i) = V^*\rho(r_i)V$$

for all $i = 1, \ldots, d$.

Note that by identifying $V\mathcal{H}$ with $\mathcal{H}$ as a closed subspace of $\mathcal{K}$, we have:

$$\pi(s_i) = P_{\mathcal{H}}\rho(r_i)\Big|_{\mathcal{H}}.$$

This shows that $\pi(s_i)$ may be viewed as a compression of $\rho(r_i)$. Moreover, by decomposing $\mathcal{K} = \mathcal{H} \oplus \mathcal{H}^\perp$, one also has:

$$\rho(r_i) = \begin{pmatrix} \pi(s_i) & \beta \\ \gamma & \delta \end{pmatrix}.$$

**Proposition 4.4.** $\mathbf{s}$ *dilates to* $\mathbf{r}$ *if and only if there exists a UCP map of* $\mathcal{S}(\mathbf{s})$ *and* $\mathcal{S}(\mathbf{r})$ *mapping* $s_i$ *to* $r_i$ *for all* $i$*.*

*Proof.* Assume that $\mathbf{s}$ dilates to $\mathbf{r}$ and choose $\pi, \rho, V$ as in 4.3. Then, writing:

$$\phi(r) = \pi^{-1}(V^*\rho(r)V)$$

gives us a UCP (as a composition of such) which maps $r_i \to s_i$. For the converse, choose some faithful representation $\pi : C^*(\mathbf{s}) \to C^*(\mathbf{s}') \subset B(\mathcal{H})$ for some Hilbert space $\mathcal{H}$. If $\phi : \mathcal{S}(\mathbf{r}) \to C^*(\mathbf{s})$ is a UCP, so is the map $\pi \circ \phi : \mathcal{S}(\mathbf{r}) \to C^*(\mathbf{s}')$. By applying Arveson's extension theorem, we may extend this map to a UCP $\Psi : C^*(\mathbf{r}) \to C^*(\mathbf{s}')$, and then apply Stinespring dilation theorem and find some Hilbert space $\mathcal{K}$, a $*$-homomorphism $\rho : C^*(\mathbf{r}) \to B(\mathcal{K})$ and an isometry $V : \mathcal{H} \to \mathcal{K}$ such that:

$$\Psi(r) = V^*\rho(r)V.$$

This means that for all $r \in \mathcal{S}(\mathbf{r})$:

$$\pi(\phi(r)) = \Psi(r) = V^*\rho(r)V$$

and by assumption, $\phi(r_i) = s_i$ so:

$$\pi(s_i) = V^*\rho(r_i)V$$

which finishes the proof. □

The following corollary uses this theorem along with Theorem 3.24.

**Corollary 4.5.** *Let* $\mathbf{s}, \mathbf{r}$ *be two d-tuples of unitaries, such that* $\mathcal{W}(\mathbf{s}), \mathcal{W}(\mathbf{r})$ *are* $N_{\mathbf{s}}$*-generated and* $N_{\mathbf{r}}$*-generated respectively, and set* $N = \max\{N_{\mathbf{s}}, N_{\mathbf{r}}\}$*. Then, the following are equivalent:*

1. $\mathbf{s}$ *and* $\mathbf{r}$ *are* $N$*-order equivalent.*
2. $\mathbf{s}$ *and* $\mathbf{r}$ *are completely order equivalent.*
3. $\mathcal{W}_N(\mathbf{s}) = \mathcal{W}_N(\mathbf{r})$*.*
4. $\mathcal{W}(\mathbf{s}) = \mathcal{W}(\mathbf{r})$*.*
5. $\mathbf{s}$ *and* $\mathbf{r}$ *are* $*$*-isomorphic.*

This corollary shows us that for unitary tuples with finitely generated matrix ranges /matrix state spaces, we can recover the $C^*$-algebraic structure by studying the $N$-order structure of some finite dimensional $N$.

Recall that $\mathcal{W}(\mathbf{s})$ being $N$-generated, means that all of the finite dimensional boundary representations of $\mathcal{S}(\mathbf{s})$ are of dimension at most $N$. Therefore, it is only natural to assume that by studying the $N$-order structure of the matrix state space, we can find algebraic information from the representations which in encoded in that set.

# Chapter 5

# Rotation Algebras and $q$-Commuting Unitaries

**Definition 5.1.** Let $\Lambda = (\lambda_{ij})_{i,j=1}^{d}$ be a self-adjoint complex matrix such that $|\lambda_{ij}| = 1$ for all $i, j$. A tuple $\mathbf{s} = (s_1, \ldots, s_d)$ of unitaries is said to be $\Lambda$-*commuting* if $s_i s_j = \lambda_{ij} s_j s_i$ for all $i, j$.

In the particular case of $d = 2$, we will say that two unitaries $u, v$ are $q$-commuting if $uv = qvu$ for some complex $q$ such that $|q| = 1$.

An operator system generated by a $\Lambda$-commuting tuple will be called a *rotation operator system* if $d = 2$ and a *higher dimensional rotation system* if $d > 2$.

So far, we saw that tuples with $N$-generated matrix ranges have the special property that their complete order structure can be recovered from their $N$-order structure. Moreover, when considering unitary tuples, the $N$-order structure completely determines the $C^*$-algebraic structure.

In this chapter, we study the matrix state space of rotation operator systems and matrix ranges of $\Lambda$-commuting tuples, and show that in some cases, the results given in Theorems 3.18 and 3.24 can be improved.

$\Lambda$-commutation relations can be described by some self adjoint $\Lambda = (\lambda_{ij})_{i,j=1}^{n} \in \mathbb{M}_n$, and a set $\mathcal{G} = \{s_1, \ldots, s_d\}$ of generators which satisfy the set:

$$\Omega = \left\{ s_i s_i^* = s_i^* s_i = 1,\ \forall i,\ \ s_i s_j = \lambda_{ij} s_j s_i,\ \forall i, j \right\}$$

of relations. It is then interesting to study the *universal* $C^*$-algebra for this set of generators and relations, which is usually denoted by $C^*(\mathcal{G}|\Omega)$. This algebra has the property that whenever $\mathbf{r} = (r_1, \ldots, r_d)$ is a $d$-tuple of $\Lambda$-commuting unitaries, there exists a $*$-homomorphism $\pi : C^*(\mathcal{G}|\Omega) \to C^*(\mathbf{r})$ mapping $s_i$ to $r_i$.

A $d$-tuple $\mathbf{r}$ is called *universal* if there exists a $*$-isomorphism of $C^*(\mathbf{r})$ and $C^*(\mathcal{G}|\Omega)$ mapping $r_i$ to $s_i$.

In this work we primarily deal with the case $d = 2$. Namely, the univeral $C^*$-algebra generated by $q$-commuting unitaries.

**Definition 5.2.** Let $\theta \in [0, 2\pi]$ and set $q = e^{i\theta}$. *The rotation algebra* $\mathcal{A}_\theta$ is the universal $C^*$-algebra generated by two $q$-commuting unitaries which will be denoted by $u_\theta, v_\theta$. We also denote by $\mathcal{S}_\theta = \mathcal{S}(u_\theta, v_\theta)$.

For the existence of a universal rotation algebra, one may refer to [6, Chapter VI].

**Examples of $q$-commuting unitaries** -

- Let $\mathcal{H} = L^2(\mathbb{T})$ and define $U, V \in B(\mathcal{H})$ by:

$$Uf(z) = zf(z), \quad Vf(z) = f(\bar{q}z).$$

  Then, $U, V$ are $q$-commuting.

- Assume that $q$ is rational. Namely, of the form $q = e^{2\pi i \frac{k}{n}}$ for coprime $k, n \in \mathbb{N}$. Define $\tilde{u}, \tilde{v} \in \mathbb{M}_n = B(\mathbb{C}^n)$ by:

$$\tilde{u} = \operatorname{diag}\big(1, q, \ldots, q^{n-1}\big), \quad \tilde{v} = \begin{pmatrix} 0 & 0 & \ldots & 0 & 1 \\ 1 & 0 & \ldots & 0 & 0 \\ 0 & 1 & \ldots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ldots & 1 & 0 \end{pmatrix} \tag{5.1}$$

  then, $\tilde{u}, \tilde{v}$ are $q$-commuting. This is sometimes called *the standard representation* for rational $q$.

- The finite dimensional case generalizes for the case of irrational $q$. Namely, $q = e^{2\pi i\theta}$ for $\theta \in [0,1] \setminus \mathbb{Q}$, by defining $U, V \in B(\ell^2(\mathbb{Z}))$ by:

$$U(x_n)_{n\in\mathbb{Z}} = (q^n x_n)_{n\in\mathbb{Z}}, \quad V(x_n)_{n\in\mathbb{Z}} = (x_{n-1})_{n\in\mathbb{Z}}$$

  this pair is again irreducible.

- Let $\Gamma(\mathcal{H}) := \bigoplus_{k=0}^{\infty} \mathcal{H}^{\otimes_s k}$ be the symmetric Fock space over a Hilbert space $\mathcal{H}$ (see [17]). The exponential vectors $e(x) := \sum_{k=0}^{\infty} \frac{1}{\sqrt{k!}} x^{\otimes k}$ for $x \in \mathcal{H}$ form a linearly independent and total subset of $\Gamma(\mathcal{H})$. A direct calculation shows that $\langle e(x), e(y)\rangle = e^{\langle x,y\rangle}$ for all $x, y \in \mathcal{H}$. For some $z \in \mathcal{H}$, define:

$$W(z)e(x) = e(z+x)e^{-\frac{\|z\|^2}{2} - \langle z,x\rangle}$$

  for all exponential vectors $e(x)$. These operators are unitaries and they satisfy:

$$W(y)W(z) = e^{2i\langle z,y\rangle} W(z)W(y),$$

  Therefore, by choosing a Hilbert space with two vectors satisfying $2\langle z, y\rangle = \theta$, we have that the two unitaries are $q = e^{i\theta}$-commuting.

Note that for irrational $q$, the algebra $\mathcal{A}_\theta$ is simple (see for example [3, Theorems 1.9 and 1.10]), which shows that any pair of $q$-commuting unitaries is universal. For the rational case, for which $q = e^{2\pi i \frac{k}{n}}$, we know that the universal algebra is not simple. This follows from the fact it has nonequivalent finite dimensional representations. An easy example of two such representations are the pairs $(\tilde{u}, \tilde{v})$ and $(\alpha\tilde{u}, \beta\tilde{v})$ where $\tilde{u}, \tilde{v}$ are the standard representation of $q$-commuting unitaries, and at least one of $\alpha, \beta$ is not of the form $q^k$ for some $k = 0, \ldots, n-1$.

In order to determine whether a pair of $q$-commuting unitaries is universal for rational $q$, we have the following lemma (see also [12, Lemma 2.2]):

**Lemma 5.3.** *Let $\mathcal{A}_\theta$ be a rational rotation algebra (so $q = e^{i\theta}$ where $\theta = 2\pi\frac{k}{n}$), and let $U, V \in B(\mathcal{H})$ be a pair of $q$-commuting unitaries. Then, the following are equivalent:*

1. *The mapping $u_\theta \mapsto U, v_\theta \mapsto V$ extends to a $*$-isomorphism of $\mathcal{A}_\theta$ and $C^*(U, V)$.*
2. *For all $\alpha, \beta \in \mathbb{T}$, the mapping $U \mapsto \alpha U, V \mapsto \beta V$ extends to a $*$-automorphism of $C^*(U, V)$.*
3. *For all $\alpha, \beta \in \mathbb{T}$, there is a representation of $C^*(U, V)$ with $U \mapsto \alpha\tilde{u}, V \mapsto \beta\tilde{v}$ where $\tilde{u}, \tilde{v}$ are given by the standard representation.*

Using this lemma, it was shown that for rational $q$, the $q$-commuting Weyl unitaries are an example of a universal pair. It is also straight forward from the lemma that each element of a universal pair must have a full spectrum in the unit circle (although it is not sufficient), which shows that (for example) the standard representation is not universal.

In this chapter, we show another significant difference between the rational and irrational $q$-commuting unitaries, which is in their matrix state space and matrix range.

As a consequence of Theorem 4.1, if $\mathbf{s}$ is a $d$-tuple $\Lambda$-commuting unitaries and $\mathbf{r}$ is a $d$-tuple of $\tilde{\Lambda}$-commuting unitaries, then the tuples are completely order equivalent if and only if they are $*$-isomorphic, and then we deduce that $\Lambda = \tilde{\Lambda}$.

For the case $d = 2$, we show that in certain cases, the complete order structure can be recovered from some $N$-order structure, and use that to classify the $*$-isomorphism equivalence classes for $q$-commuting unitaries.

## 5.1 Rational Rotation Algebras

In this section we assume that $q$ is of the form $q = e^{2\pi i\frac{k}{n}}$ for some coprime $k, n \in \mathbb{N}$. We will denote such $q$ by $q_{k,n}$ for convenience.

If $(u, v)$ is a pair of $q_{k,n}$-commuting unitaries, Theorem 3.7 gives us that absolute matrix extreme points in $\mathcal{W}(u, v)$ correspond to boundary representations of $C^*(u, v)$ for $\mathcal{S}(u, v)$. Because $u, v$ are unitaries, we have by Theorem 4.1, that any irreducible representation is a boundary representation. Therefore, the following proposition gives us a complete description of the absolute matrix extreme points in the rational case.

**Proposition 5.4.** *Let $\mathcal{A} = C^*(u, v)$ be the $C^*$-algebra generated by $q = q_{k,n}$-commuting unitaries. If $\pi : \mathcal{A} \to B(\mathcal{H})$ is irreducible, then:*

- $\dim \mathcal{H} = n$.
- *There exists an orthonormal basis and two scalars $\lambda, \zeta$ of size 1, such that $\pi(u), \pi(v)$ can be written in the form:*

$$\pi(u) = \lambda\tilde{u}, \quad \pi(v) = \zeta^{-1}\tilde{v}$$

*for some $\lambda, \zeta \in \mathbb{C}$ of modulus 1, and $\tilde{u}, \tilde{v}$ are given by (5.1).*

*Proof.* Assume that $\pi : \mathcal{A} \to B(\mathcal{H})$ is irreducible and define $U = \pi(u), V = \pi(v)$. For all $k, m \in \mathbb{N}$ we have that:

$$U^n\pi\big(u^k v^m\big) = \pi\big(u^{n+k}v^m\big) = \overbrace{q_{k,n}^{nm}}^{=1}\pi\big(u^k v^m u^n\big) = \pi\big(u^k v^m\big)U^n,$$

and the same works for $V^n$, which proves that $U^n, V^n \in \pi(\mathcal{A})'$. Irreducibility of $\pi$ means that $U^n, V^n$ must be scalar multiples of the identity, and thus, there exists some $\mu, \eta \in \mathbb{C}$ such that $|\mu| = |\eta| = 1$ and $U^n = \mu \mathbb{1}_{\mathcal{H}}, V^n = \eta \mathbb{1}_{\mathcal{H}}$. Therefore, picking any nonzero vector $\xi \in \mathcal{H}$, we have that the subspace:

$$\mathcal{K} = \operatorname{span}\{U^k V^m \xi \big| k, m \in \{-n+1, \ldots, n-1\}\} \subset \mathcal{H}$$

is a invariant for $\pi(\mathcal{A})$, so irreducibility forces $\mathcal{K} = \mathcal{H}$. This means that $\dim \mathcal{H}$ is finite, and for $U$, which is a unitary operator acting on a finite dimensional case, one may find an eigenvector $\xi \in \mathcal{H}$ of norm 1 such that $U\xi = \lambda\xi$ for some $\lambda \in \mathbb{C}, |\lambda| = 1$. Note that since $U^n = \mu \mathbb{1}_{\mathcal{H}}$, we have that $\lambda^n = \mu$. Next, we can use $\xi$ to find more eigenvectors, by using powers of $V$:

$$U\big(V^k \xi\big) = q^k V^k (U\xi) = \lambda q^k V^k \xi$$

which means that $q^k \lambda$ is another eigenvalue for $U$. This gives us another subspace:

$$\mathcal{K} = \operatorname{span}\{\xi, V\xi, \ldots, V^{n-1}\xi\}$$

which is invariant for $\pi(\mathcal{A})$ and thus $\mathcal{K} = \mathcal{H}$ and $\dim \mathcal{H} = n$. Lastly, choose some $\zeta \in \mathbb{C}$ such that $\zeta^n = \eta^{-1}$. The basis $\{\zeta^{i-1} V^{i-1} \xi\}_{i=1}^n$ is an orthonormal basis of eigenvectors for $U$, on which:

$$V\big(\zeta^{i-1} V^{i-1} \xi\big) = \zeta^{i-1} V^i \xi = \zeta^{-1}\big(\zeta^i V^i \xi\big)$$

for $i = 1, \ldots, n-1$ and $V(\zeta^{n-1} V^{n-1} \xi) = \eta \zeta^{n-1} \xi = \zeta^{-1} \xi$. Therefore $U$ and $V$ take the form:

$$U = \lambda \operatorname{diag}\big(1, q, \ldots, q^{n-1}\big), \quad V = \zeta^{-1} \begin{pmatrix} 0 & 0 & \ldots & 1 \\ 1 & 0 & \ldots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \ldots & 1 & 0 \end{pmatrix}$$

as we wanted to show. □

Applying the proposition with our description of the absolute matrix extreme points in $\mathcal{W}(\mathbf{s})$ yields the following corollary:

**Corollary 5.5.** *Let $\mathcal{S} = \mathcal{S}(u, v)$ be an operator system generated by $q = q_{k,n}$-commuting unitaries. Then* $\operatorname{abex}(\operatorname{UCP}(\mathcal{S}))$ *is not empty, and is contained in* $\operatorname{UCP}(\mathcal{S}, \mathbb{M}_n)$.

*Proof.* The results of Davidson and Kennedy [8] show that any operator system admits boundary representations. By Proposition 5.4, they all lie within $\operatorname{UCP}(\mathcal{S}(u, v), \mathbb{M}_n)$, and they are all absolute matrix extreme points by Theorem 3.7. By the same theorem, any absolute matrix extreme point arises this way, which completes the proof. □

Using our characterization of absolute matrix extreme points and matrix extreme points in $\operatorname{UCP}(\mathcal{S})$, we now have a detailed description of the matrix state space for operator systems generated by $q_{k,n}$-commuting unitaries.

**Theorem 5.6.** *Let $\mathcal{S} = \mathcal{S}(u, v)$ be an operator system generated by $q_{k,n}$-commuting unitaries. Then:*

1. $\operatorname{UCP}(\mathcal{S})$ *is $n$-generated.*

2. $\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n) = \mathrm{abex}(\mathrm{UCP}(\mathcal{S}))$ *and this is the set of all boundary representations for* $\mathcal{S}$.

3. *for* $1 \leq k < n$, $\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_k)$ *is non-empty, and any* $\varphi \in \partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_k)$ *is of the form* $\varphi = \gamma^* \psi \gamma$ *for some* $\psi \in \partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$.

*Proof.* We start by proving that if $\varphi \in \partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_m)$, then $m \leq n$. Indeed, Theorem 3.5 shows that $\varphi$ being matrix extreme implies it is pure. Therefore, we may invoke [8, Theorem 2.4], and get that $\varphi$ can be written as $\varphi = V^* \pi|_{\mathcal{S}} V$, where $\pi$ is a boundary representation for $\mathcal{S}$, and $V$ is an isometry. By Proposition 5.4, $\pi|_{\mathcal{S}} \in \mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$, from which we deduce that $m \leq n$.

Next, we use Corollary 5.5 and Theorem 3.7, to conclude that $\emptyset \neq \mathrm{abex}(\mathrm{UCP}(\mathcal{S})) \subset \mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$, which proves (1).

For the second point, note that we have the inclusion $\mathrm{abex}(\mathrm{UCP}(\mathcal{S})) \subset \partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$ by Corollary 5.5. For the other inclusion, note that if $\varphi \in \partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$, it is either an absolute matrix extreme point or it dilates to a matrix extreme point in $\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_{n+1})$, by Proposition 3.9. Since we proved that $\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_{n+1})$ is empty, $\varphi$ has to be an absolute matrix extreme point.

As for the third point, we use that fact that $\mathrm{UCP}(\mathcal{S}, \mathbb{C})$ satisfies all of the condition of the Krein-Milman theorem, and thus $\mathrm{ext}(\mathrm{UCP}(\mathcal{S}, \mathbb{C}))$ is non-empty. Any extreme point of this set must be trivially matrix extreme, because it is not a proper matrix convex combination of points of lower levels. By Proposition 5.5 it does not extend to a boundary representation, so by Theorem 3.7 and Proposition 3.9, it is a compression of a matrix extreme point in $\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_2)$. In particular, this set is not empty.

Using induction on levels below $n$, we can continue this process and find matrix extreme points in all level below $n$, all of the are not matrix extreme. Finally, because these are pure UCP maps which are not maximal, they have by [8, Theorem 2.4] a dilation to a boundary representation, which is some $\psi \in \partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_n)$, and this proves (3). □

The next theorem shows that for the class of rational $q$-commuting unitaries, we can improve the results of Theorem 3.24 significantly. Even though this theorem can be deduced from Theorem 5.15, we leave the proof here because the techniques in this proof might be used for other classes of finitely generated matrix state spaces.

**Theorem 5.7.** *Let* $u, v$ *be a pair of* $q = q_{k,n}$*-commuting unitaries,* $u', v'$ *be a pair of* $q' = q_{k',n'}$*-commuting unitaries, and set* $N = \min\{n, n'\} + 1$. *Then,* $(u, v)$ *and* $(u', v')$ *are* $N$*-order equivalent if and only if the tuples are* $*$*-isomorphic.*

*Proof.* Assume by contradiction that $n \neq n'$, and without loss of generality, assume that $n < n'$. We apply Theorem 5.6 to deduce that $\mathrm{UCP}(\mathcal{S}(u', v'), n+1)$ is non-empty, so by Proposition 3.16, we get that $\partial\mathrm{UCP}(\mathcal{S}(u, v), n+1)$ is non-empty as well. This is a contradiction to Theorem 5.6, so $n = n'$. Thus, we get that $N \geq \max\{n, n'\}$, so we may invoke Corollary 4.5 to conclude that the pairs are $*$-isomorphic. In particular, we get that $q = q'$. □

This theorem shows that in these cases of $q$-commuting unitaries for rational $q$'s, studying the $N$-order structure in the matrix state space (for some $N$ which depends on $q$), is sufficient to determine the complete order structure, and because we are discussing unitary tuples, this also determines the $C^*$-algebraic structure generated by such tuples.

A natural question to ask is whether any two $q_{k,n}$-commuting pairs are $n$-order isomorphic. The example below shows that it need not be the case. In fact, this example is true for all rational $q_{k,n}$.

**Example 5.8.** Let $q = e^{2\pi i \frac{k}{n}}$ such that $\gcd(k,n) = 1$.

- Let $u, v \in B(L^2(\mathbb{T}))$ be the operators:
$$uf(z) = zf(z), \quad vf(z) = f(\bar{q}z)$$
it is evident that $u, v$ are unitaries which satisfy $uv = qvu$ (see [6, Chapter 6]). It is also evident that $C^*(u)$ is isomorphic to $C(\mathbb{T})$, a fact we will need later on.
- Let $\tilde{u}, \tilde{v}$ be the standard representation of $q_{k,n}$-commuting unitaries.

Those are two $q_{k,n}$-commuting pairs, and we will show that they are not $n$-order equivalent. Since the two pairs have $n$-generated matrix ranges (by Theorem 5.6), we get by Corollary 4.5 that if we assumed that they are $n$-order equivalent, they would also be $*$-isomorphic. Note that $\sigma(u) = \mathbb{T}$ and $\sigma(\tilde{u})$ is finite, which contradicts the pairs being $*$-isomorphic.

Recall that in the rational case, the only irreducible representations of $q_{k,n}$-commuting unitaries are of the form $\alpha\tilde{u}, \beta\tilde{v}$ where $\tilde{u}, \tilde{v}$ is the standard representation given in (5.1) and $\alpha, \beta \in \mathbb{T}$. We can now use the new tools developed to find the equivalence classes of the irreducible representations.

**Corollary 5.9.** *Let $\mathcal{A}_\theta$ be the rational rotation algebra for $\theta = 2\pi\frac{k}{n}$ with coprime $k, n \in \mathbb{N}$. Then two irreducible representations $\pi : \mathcal{A}_\theta \to B(\mathcal{H})$ and $\rho : \mathcal{A}_\theta \to B(\mathcal{K})$ are unitarily equivalent if and only if $(\pi(u_\theta), \pi(v_\theta))$ is $n$-order equivalent to $(\rho(u_\theta), \rho(v_\theta))$.*

*Proof.* By Proposition 5.4, we know that $(\pi(u_\theta), \pi(v_\theta))$ are unitarily equivalent to $(\alpha\tilde{u}, \beta\tilde{v})$ for some $\alpha, \beta \in \mathbb{T}$ and that $(\rho(u_\theta), \rho(v_\theta))$ are unitarily equivalent to $(\alpha'\tilde{u}, \beta'\tilde{v})$. By Corollary 4.5, the tuples are $n$-order equivalent if and only if they are $*$-isomorphic, which means that they have the same spectrum. Note that:
$$\sigma(\alpha\tilde{u}) = \alpha\sigma(\tilde{u}), \quad \sigma(\beta\tilde{v}) = \beta\sigma(\tilde{v})$$
and the same is true for the second pair. Since both operators have $n$ distinct spectrum points, the $*$-isomorphism is possible only if $\alpha' = \alpha q^\ell$ and $\beta' = \beta q^m$ for some $\ell, m \in \mathbb{Z}$. This means that:
$$\rho(u_\theta) \cong_{\text{Uni}} \alpha\text{diag}\big(q^\ell, q^{\ell+1}, \ldots, q^{\ell+n-1}\big), \quad \rho(v_\theta) \cong_{\text{Uni}} \beta q^m \tilde{v},$$
and therefore,shifting the order of the basis to the right $\ell - m$ gives a unitary equivalence between this pair and $(\alpha q^m\tilde{u}, \beta q^m\tilde{v})$ which is unitarily equivalent to $(\pi(u_\theta), \pi(v_\theta))$. □

Note that for representations which are not irreducible, this corollary need not be true. The example below shows the corollary will not work even in finite dimensional cases.

**Example 5.10.** Let $\tilde{u}, \tilde{v}$ be the standard representation of $q_{k,n}$-commuting unitaries, and let $u = \tilde{u} \oplus \tilde{u}, v = \tilde{v} \oplus \tilde{v}$. Then, the tuples are obviously not unitarily equivalent (for example, because the dimensions of the spaces they act on are different), but they are $*$-isomorphic and therefore $n$-order equivalent.

The corollary fails even harder when dealing with irrational $q$-commuting unitaries, and we will show that there are irreducible representations which are not unitarily equivalent.

## 5.2 Irrational rotation algebras

In this section, we say that $q \in \mathbb{T}$ is irrational if it is of the form $q = e^{i\theta}$ for $\theta \in [0, 2\pi] \setminus 2\pi\mathbb{Q}$. The first known result we will state, is that for $q$-commuting unitaries with irrational $q$s have no finite dimensional representations.

**Proposition 5.11.** *Let $u, v$ be $q$-commuting unitaries acting on some Hilbert space $\mathcal{H}$ such that $q$ is irrational. Then $\mathcal{H}$ is infinite dimensional.*

*Proof.* Assume by contradiction that $\dim \mathcal{H} = N$ for some $N$. Then, $u$ is a unitary acting on a finite dimensional space and thus has an eigenvector $\xi \in \mathcal{H}$ for a non-zero eigenvalue $\lambda \in \mathbb{C}$. Note that since $v$ is unitary, we get for all $k \in \mathbb{Z}$:

$$u\big(v^k\xi\big) = q^k v^k u\xi = \lambda q^k \big(v^k\xi\big).$$

$q$ being irrational, means that $\big(v^k\xi\big)_{k\in\mathbb{Z}}$ are all eigenvectors of different eigenvalues, which is a contradiction to $\dim \mathcal{H}$ being finite dimensional. □

Another property of irrational $q$-commuting unitaries, is that they all generate isomorphic $C^*$-algebras.

**Theorem 5.12.** *[6, Theorem 4.1.4] Let $q \in \mathbb{T}$ be irrational and denote by $\mathcal{A}_q$ the universal $C^*$-algebra generated by $q$-commuting unitaries. Then, if $u, v$ are any other $q$-commuting unitary pair, then $C^*(u, v)$ is $*$-isomorphic to $\mathcal{A}_q$.*

**Theorem 5.13.** *Let $\mathcal{S} = \mathcal{S}(u, v)$ be an operator system generated by $q$-commuting unitaries, with $q$ being irrational. Then:*

1. $\mathrm{UCP}(\mathcal{S})$ *is infinitely generated.*
2. $\mathrm{UCP}(\mathcal{S})$ *has no absolute matrix extreme points.*
3. $\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_k)$ *is non empty for all $k \in \mathbb{N}$, and they are all of the form $V^*\,\pi\big|_{\mathcal{S}}\,V$ for some boundary representation $\pi$ for $\mathcal{S}$.*

*Proof.* We start with the fact that by Theorem 3.7, and by Proposition 5.11, the absence of finite dimensional representation (in particular, boundary representations), implies that $\mathrm{abex}(\mathrm{UCP}(\mathcal{S}))$ is empty, which proves (2).

Using the same technique as in the rational case, we use the fact that $\mathrm{UCP}(\mathcal{S}, \mathbb{C})$ has an extreme point (as a compact convex set in a locally convex vector space), which has to be matrix extreme. Then, we apply Proposition 3.9, an get that the extreme point in level 1 dilates to another matrix extreme point in level 2 which means that $\partial\mathrm{UCP}(\mathcal{S}, \mathbb{M}_2)$ is non-empty. Again, the matrix extreme points in this level cannot be absolute because they are not boundary representations, which means we can continue inductively and fine matrix extreme points at every level of $\mathrm{UCP}(\mathcal{S})$. This proves (1), and (3) is proven by also stating that each matrix extreme point is a pure UCP map and by [8, Theorem 2.4], is a compression of a boundary representation (in this case, on an infinite dimensional Hilbert space). □

We see that in the irrational case, our previous methods do not get us below the complete order structure. Although, using this characterization we can still improve Theorem 5.7 to include a case for irrational commuting pairs.

**Corollary 5.14.** *Let $(u, v)$ be $q_{k,n}$-commuting unitaries and $(u', v')$ be a pair of $q$-commuting unitaries, where $q$ is irrational. Then, the tuples are not $(n+1)$-order equivalent.*

*Proof.* Again, by applying Proposition 3.16, the existence of such $(n+1)$-order isomorphism would imply that $\partial\mathrm{UCP}\left(\mathcal{S}(u, v), \mathbb{M}_{n+1}\right)$ is non-empty because $\partial\mathrm{UCP}\left(\mathcal{S}(u', v'), \mathbb{M}_{n+1}\right)$ is non-empty (by Theorem 5.13), and this is impossible by Theorem 5.6. □

### 5.2.1 Equivalence in the class of $q$-commutation

We already know by Theorem 5.7, that for rational $q_{k,n}$-commuting unitaries and $q_{k',n'}$-commuting unitaries, the existence of a $(\min\{n, n'\}+1)$-order isomorphism will imply complete order isomorphism (and thus $*$-isomorphism of the corresponding algebras, by Theorem 4.2). Therefore, in this case, we get that $q_{k,n} = q_{k',n'}$.

This result takes obvious advantage of the fact that the matrix state spaces are finitely generated, which means that the constant $q$ gives us some information regarding the complete order equivalence class of tuples of $q$-commuting unitaries.

One would hope that $\min\{n, n'\}$-order isomorphism would be sufficient. This turns out to be the case, and the proof uses a different approach by considering inclusions of matrix ranges and the relation of those inclusions to dilation problems.

**Theorem 5.15.** *Let $u, v$ be $q_{k,n}$-commuting unitaries and $u', v'$ are $q$-commuting unitaries for $q \in \mathbb{T}$. Assume that the mapping $u' \mapsto u$ and $v' \mapsto v$ defines a unital and $n$-positive map. Then, $q = q_{k,n}$. In particular, if the mapping is an $n$-order isomorphism, the pairs are completely order isomorphic and generate isomorphic $C^*$-algebras.*

*Proof.* By applying Theorem 2.11, we have that for all $k \leq n$:

$$\mathcal{W}_k(u, v) \subset \mathcal{W}_k(u', v').$$

Next, we use Proposition 3.23, which shows that $\mathcal{W}(u, v)$ is the minimal matrix convex set which contains $\mathcal{W}_n(u, v)$ which means we now have the inclusion $\mathcal{W}(u, v) \subset \mathcal{W}(u', v')$. Again by Theorem 2.11, the inclusion of the matrix ranges implies that the mapping is in fact UCP.

In Theorem 4.1, we showed that a UCP map which maps a tuples of unitaries to a tuple of unitaries extends to a $*$-homomorphism of the generated $C^*$-algebras, which we denote by $\pi : C^*(u', v') \to C^*(u, v)$. $\pi$ being multiplicative, implies:

$$\pi(u'v') = \pi(u')\pi(v') = uv = q_{k,n}vu$$

and:

$$\pi(u'v') = \pi(qv'u') = q\pi(v')\pi(u') = qvu$$

from which it follows that $q = q_{k,n}$. In particular, if the mapping is an $n$-order isomorphism, we can now invoke Corollary 4.5 and finish the proof. □

## 5.3 Rational $\Lambda$-commuting $d$-tuples

In this section we consider $d$-tuples of $\Lambda$-commuting unitaries. We say that $\Lambda$ is rational if $\lambda_{ij} = e^{2\pi i \frac{k_{ij}}{n_{ij}}}$ for all $i, j$. Just as in Proposition 5.4, we can give a bound on the dimension of irreducible representations of rational $\Lambda$-commuting unitaries.

**Proposition 5.16.** *Let* $\mathbf{s}$ *be a* $d$*-tuple of* $\Lambda$*-commuting unitaries for some rational* $\Lambda$*, and set* $N = \operatorname{lcm}\{n_{ij}\}_{i,j}$*. If* $\pi : C^*(\mathbf{s}) \to B(\mathcal{H})$ *is irreducible, then* $\mathcal{H}$ *is finite dimensional, and* $\dim \mathcal{H} \leq N^d$*.*

*Proof.* Denote $U_i = \pi(s_i)$ for all $i$, and note that:

$$U_i^N \prod_{j=1}^{d} U_j^{r_j} = \left( \prod_{\substack{j=1 \\ j \neq i}}^{d} \lambda_{ij}^{N r_j} \right) \left( \prod_{j=1}^{k} U_j^{r_j} \right) U_i^N = \left( \prod_{j=1}^{k} U_j^{r_j} \right) U_i^N$$

which means that $U_i^N \in \pi(C^*(\mathbf{s}))'$ and thus must be a scalar multiple of the identity for all $i = 1, \ldots, d$. Therefore, choosing any non-zero vector $\xi \in \mathcal{H}$, we have that the subspace:

$$\tilde{\mathcal{H}} = \operatorname{span} \left\{ \prod_{j=1}^{d} U_i^{r_j} \xi \,\middle|\, r_j \in \{0, \ldots, N-1\} \, \forall j \right\}$$

is a reducing finite dimensional subspace of $\mathcal{H}$. By irreducibility, it must hold that $\tilde{\mathcal{H}} = \mathcal{H}$ which completes the proof. □

It is worth noting that $N^d$ is just an upper bound, and there are cases of representations on lower dimensional spaces (see Proposition 5.4). Another example is given for $d = 3$:

**Proposition 5.17.** *[13, Proposition B.1] Let* $q = e^{2\pi i \frac{k}{n}}$ *for coprime* $k, n \in \mathbb{N}$*, and let* $(u, v, w)$ *be a tuple of irreducible* $\Lambda$*-commuting unitaries in* $B(\mathcal{H})$*, such that:*

$$\Lambda = \begin{pmatrix} 1 & q & \bar{q} \\ \bar{q} & 1 & q \\ q & \bar{q} & 1 \end{pmatrix}.$$

*Then,* $\dim \mathcal{H} = n$*, and there exists some orthonormal basis such that:*

$$u = \alpha \tilde{u}, \quad v = \beta \tilde{v}, \quad w = \gamma \tilde{u} \tilde{v}$$

*where* $\tilde{u}, \tilde{v}$ *are given by the standard representation in (5.1).*

Unlike the example given in this proposition, we do not now whether all irreducible representations of a rational $\Lambda$-commuting unitaries have the same dimension. But by the universal property, we know that any irreducible representation is a representation of the universal pair. This motivates the following:

**Proposition 5.18.** *Let* $\Lambda = (\lambda_{ij})_{i,j=1}^{d}$ *be rational and let* $\mathbf{s}^\Lambda$ *be the universal* $d$*-tuple of* $\Lambda$*-commuting unitaries. Then if* $\operatorname{UCP}\left(\mathcal{S}(\mathbf{s}^\Lambda)\right)$ *is* $N(\Lambda)$*-generated, then whenever* $\mathbf{r}$ *is a* $d$*-tuple of* $\Lambda$*-commuting unitaries, then* $\operatorname{UCP}\left(\mathcal{S}(\mathbf{r})\right)$ *is* $n$*-generated for some* $n \leq N(\Lambda)$*.*

*Proof.* By [8, Theorem 4.2], we know that if $\varphi \in \mathrm{UCP}\left(\mathcal{S}(\mathbf{r}), \mathbb{M}_n\right)$ is matrix extreme, it dilates to a restriction of a boundary representation for $\mathcal{S}$. Therefore, it is sufficient to prove that if $\pi : C^*(\mathbf{r}) \to B(\mathcal{H})$ is a boundary representation for $\mathcal{S}(\mathbf{r})$, then $\dim \mathcal{H} \leq N(\Lambda)$. Denoting by $U_i = \pi(r_i)$ for all $i$, we have that the tuple $\mathbf{U} = (U_1, \ldots, U_d)$ is a $d$-tuple of $\Lambda$-commuting unitaries. By the universal property, there exists a $*$-homomorphism $\rho : C^*(\mathbf{s}^\Lambda) \to C^*(\mathbf{U})$, mapping $\mathbf{s}^\Lambda$ to $\mathbf{U}$. Using Theorem 4.1, we see that the restriction of this map to $\mathcal{S}(\mathbf{s}^\Lambda)$ has the unique extension property, which means that $\rho$ is a boundary representation for $\mathcal{S}(\mathbf{s}^\Lambda)$. By Theorem 3.7, it as an absolute matrix extreme point of $\mathrm{UCP}\left(\mathcal{S}(\mathbf{s}^\Lambda)\right)$, from which it follows that $\dim \mathcal{H} \leq N(\Lambda)$. □

The following statements are generalizations of some of the results that were formulated for $q$-commuting pairs. Note that sine $\mathbf{s}^\Lambda$ is a tuple of $\Lambda$ commuting unitaries (for a rational $\Lambda$), then it is $N(\Lambda) \leq N^d$ generated for some $N \in \mathbb{N}$, by using Proposition 5.16 and the characteriztion of finitely generated matrix convex sets in Chapter 3. Therefore, the $N(\Lambda)$ in the proposition does exist and also denotes the integer such that $\mathrm{UCP}\left(\mathcal{S}(\mathbf{s}^\Lambda)\right)$ is $N(\Lambda)$-generated.

**Proposition 5.19.** *Let $\Lambda = (\lambda_{ij})_{i,j=1}^d$ be rational and let $\mathbf{s}$ be a $d$-tuple of $\Lambda$-commuting unitaries. Then $\mathcal{W}(\mathbf{s}) = \mathcal{W}^{N(\Lambda)\text{-}min}(\mathbf{s})$.*

The proposition follows from Propositions 5.18 and 3.13, and allows us to deduce a $\Lambda$-commutation variation of Theorem 5.15 and Corollary 4.5.

**Theorem 5.20.** *Let $\Lambda$ be rational and let $\mathbf{s}$ be a $d$-tuple of $\Lambda$-commuting unitaries. If $\mathbf{r}$ is a $d$-tuple of $\tilde{\Lambda}$-commuting unitaries, and the mapping $r_i \mapsto s_i$ extends to a unital $N(\Lambda)$-positive map, then $\tilde{\Lambda} = \Lambda$. In particular, if the mapping is an $N(\Lambda)$-order isomorphism, the tuples are completely order isomorphic and generate isomorphic $C^*$-algebras.*

*Proof.* Under the assumptions, and by using Theorem 2.11, we get that $\mathcal{W}_{N(\Lambda)}(\mathbf{s}) \subset \mathcal{W}_{N(\Lambda)}(\mathbf{r})$. By Proposition 5.19, we get that $\mathcal{W}(\mathbf{s})$ is $N(\Lambda)$-minimal, so that the inclusion implies that the mapping $r_i \mapsto s_i$ extend to a UCP map of the generated operator systems. By Theorem 4.1, this mapping extends to a $*$-homomorphism $\pi : C^*(\mathbf{r}) \to C^*(\mathbf{s})$, so:

$$\pi(r_i r_j) = \pi(r_i)\pi(r_j) = s_i s_j = \lambda_{ij} s_j s_i$$

$$\pi(r_i r_j) = \pi\Big(\tilde{\lambda}_{ij} r_j r_i\Big) = \tilde{\lambda}_{ij} s_j s_i$$

from which it follows that $\Lambda = \tilde{\Lambda}$. Assuming that the mapping is an $N(\Lambda)$-order isomorphism, we use the same argument for the inverse map and get a complete order isomorphism, so the final part of the proof follows from Theorem 4.2. □

**Theorem 5.21.** *Let $\Lambda = (\lambda_{ij})_{i,j=1}^d$ be rational and let $\mathbf{s}$ be a $d$-tuple of $\Lambda$-commuting unitaries. For a $d$-tuple $\mathbf{r}$ of $\tilde{\Lambda}$-commuting unitaries, the following are equivalent:*

1. *$\mathbf{s}$ and $\mathbf{r}$ are $N(\Lambda)$-order equivalent.*
2. *$\mathbf{s}$ and $\mathbf{r}$ are completely order equivalent.*
3. *$\mathcal{W}_{N(\Lambda)}(\mathbf{s}) = \mathcal{W}_{N(\Lambda)}(\mathbf{r})$.*

*4.* $\mathcal{W}(\mathbf{s}) = \mathcal{W}(\mathbf{r})$.

*5.* $\mathbf{s}$ *and* $\mathbf{r}$ *are* $*$*-isomorphic.*

*Proof.* (2) $\implies$ (1) is immediate. (1) $\implies$ (3) follows 2.11. In order to prove that (3) $\implies$ (4), we use the fact that $\mathcal{W}(\mathbf{s})$ is $N(\Lambda)$-minimal, so the equality in (3) implies that $\mathcal{W}(\mathbf{s}) \subset \mathcal{W}(\mathbf{r})$. Now, Theorem 2.11 implies the existence of a UCP map of the generated operator systems mapping $r_i$ to $s_i$, which is in particular unital and $N(\Lambda)$-positive. By Theorem 5.20, we get that $\tilde{\Lambda} = \Lambda$, so that $\mathcal{W}(\mathbf{r})$ is also $N(\Lambda)$-minimal, from which we get the second inclusion and thus equality. In order to show (4) $\implies$ (2), we apply Theorem 2.11, which means there exists a unital and completely isometric map of the operator systems which maps $r_i$ to $s_i$. A unital isometry is positive, which means that this is a complete order isomorphism. Finally, (4) $\iff$ (2) and (5) $\iff$ (2) follows Theorem 4.2. □

Note that the case $d = 2$ is simpler because we already proved in Theorem 5.6, that for $q_{k,n}$-commuting unitaries, we have $N(\Lambda) = n$.

## 5.4 The isometric case

In the proof of Theorem 5.15, we saw that if a $q$-commuting pair dilates to a $q'$-commuting pair, it must hold that $q = q'$. The case $q = 1$, implies that if a pair of $q$-commuting unitaries dilate to a pair of commuting unitaries, then $q = 1$.

We shall make use of this in order to try and answer the question - could $N$-generated matrix convex sets be $m$-order isomorphic for some $m < N$ but not be $N$-order isomorphic?

We begin by referring to the following theorem:

**Theorem 5.22.** *[12, Theorem 6.4] Let* $\theta, \theta' \in \mathbb{R}, q = e^{i\theta}, q' = e^{i\theta'}$ *and* $\gamma = \theta - \theta'$. *The smallest constant* $c_{\theta,\theta'}$ *such that every pair of* $q$*-commuting unitaries can be dilated to* $c_{\theta,\theta'}$ *times a pair of* $q'$*-commuting unitaries is given by:*

$$c_{\theta,\theta'} = c_\gamma = \frac{4}{\left\|u_\gamma + u_\gamma^* + v_\gamma + v_\gamma^*\right\|},$$

*where* $u_\gamma, v_\gamma$ *are the universal pair of* $e^{i\gamma}$*-commuting unitaries.*

Applying the previous comment yields the following:

**Corollary 5.23.** *For* $\theta \in \mathbb{R}$, $c_\theta = 1$ *if and only if* $\theta \in 2\pi\mathbb{Z}$.

*Proof.* If $\theta \in 2\pi\mathbb{Z}$, there is nothing to prove, since every pair of commuting unitaries is a dilation of itself. Now assume that $\theta \notin 2\pi\mathbb{Z}$, which means that $q \neq 1$. As we stated, no pair of $q$-commuting unitaries can be dilated to a pair of commuting unitaries. From this, it must hold that $c_\theta \neq 1$. □

Referring to a numerical estimate of $c_\theta$ in [12, Figure 1], we see that there are some cases of two different values of $q$ which correspond to the same dilation constant. For example, for every $\theta \in [0, 2\pi]$, $c_\theta = c_{2\pi-\theta}$ (which besides being evident numerically, we also prove in Proposition 5.24).

Considering Theorem 5.22, we know that if the mapping $u_\theta \mapsto u_{\theta'}, v_\theta \mapsto v_{\theta'}$ extends to a unital isometry of the corresponding operator systems, then $c_\theta = c_{\theta'}$.

The following shows that the converse also holds when $\theta' = 2\pi - \theta$.

**Proposition 5.24.** *Let $\theta \in [0, 2\pi]$. Then the mapping $u_\theta \mapsto u_{2\pi-\theta}, v_\theta \mapsto v_{2\pi-\theta}$ extends to a unital isometry of the operator systems. In particular, $c_\theta = c_{2\pi-\theta}$.*

*Proof.* Assume first that $\theta = 2\pi\frac{k}{n}$ for some coprime $k, n \in \mathbb{N}$, and set $q = e^{i\theta}$. Consider the standard representation $\tilde{u}, \tilde{v}$, which we remind the reader that is given by:

$$\tilde{u} = \operatorname{diag}\big(1, q, \ldots, q^{n-1}\big), \quad \tilde{v} = \begin{pmatrix} 0 & 0 & \ldots & 0 & 1 \\ 1 & 0 & \ldots & 0 & 0 \\ 0 & 1 & \ldots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ldots & 1 & 0 \end{pmatrix}.$$

The pair $(\tilde{u}^t, \tilde{v}^t) = (\tilde{u}, \tilde{v}^t)$ is a $\bar{q} = e^{i(2\pi-\theta)}$-commuting pair, as the following calculation shows:

$$\tilde{u}^t\tilde{v}^t = (\tilde{v}\tilde{u})^t = (\bar{q}\tilde{u}\tilde{v})^t = \bar{q}\tilde{v}^t\tilde{u}^t.$$

The mapping on $\mathbb{M}_n$ given by $a \mapsto a^t$ is a unital isometry, because:

$$\big\|a^t\big\| = \sup_{\|\xi\|=1} \sqrt{\bar{\xi}^t \bar{a} a^t \xi} = \sup_{\|\xi\|=1} \sqrt{\xi^t a \bar{a}^t \bar{\xi}} = \|a\|$$

which means that its restriction to the operator system $\mathcal{S}(\tilde{u}, \tilde{v})$ is a unital isometry which maps the $q$-commuting pair $(\tilde{u}, \tilde{v})$ to the $\bar{q}$-commuting pair $(\tilde{u}, \tilde{v}^t)$.

Let $(u_\theta, v_\theta)$ be the universal pair of $q$-commuting unitaries given by:

$$u_\theta = \bigoplus_{\alpha,\beta\in\mathbb{T}} \tilde{u}_{\alpha,\beta}, \quad v_\theta = \bigoplus_{\alpha,\beta\in\mathbb{T}} \tilde{v}_{\alpha,\beta}$$

such that $\tilde{u}_{\alpha,\beta} = \alpha\tilde{u}, \tilde{v}_{\alpha,\beta} = \beta\tilde{v}$ for all $\alpha, \beta \in \mathbb{T}$. Our calculation shows that the mapping:

$$u_\theta \mapsto u_\theta, \quad v_\theta \mapsto \bigoplus_{\alpha,\beta\in\mathbb{T}} \tilde{v}^t_{\alpha,\beta}$$

extends to a unital isometry of the corresponding operator systems. From this it follows that $c_\theta = c_{2\pi-\theta}$.

Now assume that $\theta \in [0, 2\pi]\backslash 2\pi\mathbb{Q}$, and consider the infinite matrices $u = (u_{nm})_{n,m\in\mathbb{Z}}$ and $v = (v_{nm})_{n,m\in\mathbb{Z}}$ such that:

$$u_{nm} = q^n\delta_{nm}, \quad v_{nm} = \delta_{n(m+1)}$$

for all $n, m \in \mathbb{Z}$. These matrices give rise to $q$-commuting unitaries on $B(\ell_2(\mathbb{Z}))$ by ordinary matrix multiplication from the left (when we consider points as infinite column vectors).

With respect to the standard basis of $\ell_2(\mathbb{Z})$, we associate each operator $a \in B(\ell_2(\mathbb{Z}))$ with the matrix $(a_{nm})_{n,m\in\mathbb{Z}}$ given by $a_{nm} = \langle ae_m, e_n\rangle$ for all $n, m \in \mathbb{Z}$. In this case, the transposition map $(a_{nm})_{n,m\in\mathbb{Z}} \mapsto (a_{mn})_{n,m\in\mathbb{Z}}$ is again a unital isometry of $B(\ell_2(\mathbb{Z}))$ which restricts to one on $\mathcal{S}(u, v)$, and maps $(u, v)$ to the pair $(u, v^t)$ ($v^t$ being the

operator whose infinite matrix is given by $(v_{nm})^t_{n,m\in\mathbb{Z}})$, which is a pair of $\bar{q}$-commuting unitaries.

We once again define:

$$u_\theta = \bigoplus_{\alpha,\beta\in\mathbb{T}} u_{\alpha,\beta}, \quad v = \bigoplus_{\alpha,\beta\in\mathbb{T}} v_{\alpha,\beta}$$

where $u_{\alpha,\beta} = \alpha u, v_{\alpha,\beta} = \beta v$. Note that the tuple is clearly universal by Lemma 5.3, so the mapping:

$$u_\theta \mapsto u_\theta, \quad v_\theta \mapsto \bigoplus_{\alpha,\beta\in\mathbb{T}} v^t_{\alpha,\beta}$$

extends to a unital isometry of the corresponding operator systems. Once again, it follows that $c_\theta = c_{2\pi-\theta}$. □

Note that by choosing some rational $\theta = 2\pi\frac{k}{n}$, we have many examples of 1-order isomorphic tuples which are not $n$-order isomorphic.

# Chapter 6

# Concluding remarks

## 6.1 Compact matrix convex sets

Organizing and further investigating previous results, both in matrix convexity and the boundary theory, we see that if $K = (K_n)_{n=1}^{\infty}$ is a compact matrix convex set, there are precisely two kinds of matrix extreme points. The first kind is a "regular" matrix extreme point. Those are points which always dilate to another matrix extreme point at the next level of the set. These points also correspond to a pure UCP map from the operator $A(K)$ to $\mathbb{M}_n$ for some $n$. The second kind of matrix extreme points are the absolute matrix extreme points. Those are the matrix extreme points which do not dilate to matrix extreme points on higher levels, and they correspond to boundary representations of $C^*(A(K))$ for $A(K)$.

This characterization proved to be useful in understanding the structure of the compact matrix convex sets. For example, we showed in Chapter 5 that for rational $q$-commuting unitaries, all absolute matrix extreme points are at level $n$ for some $n$ which depends on $q$, and by dilating matrix extreme points from level 1 of the matrix state space, we concluded that $\mathrm{UCP}(\mathcal{S})$ has matrix extreme points at any level below $n$. This sort of phenomenon raises questions, such as:

1. Let $v_n \in K_n$ be a matrix extreme point. Is it necessarily possible to find a sequence $v_1, \ldots, v_{n-1}$ of matrix extreme points such that $v_i \in K_i$ and $v_i$ dilated to $v_{i+1}$ for $i = 1, \ldots, n-1$?

2. Let $\mathcal{S}$ be an operator systems such that $\mathrm{UCP}(\mathcal{S})$ is $n$-generated. Is it then true that all boundary representations of $\mathcal{S}$ are finite dimensional?

If the first question has a positive answer, we can use it to prove an improvement of Theorem 3.18 by replacing $N = \max\{N_{\mathcal{S}}, N_{\mathcal{R}}\}$ with $N = \min\{N_{\mathcal{S}}, N_{\mathcal{R}}\} + 1$, which is significant.

## 6.2 Representations of $\Lambda$-commuting $d$-tuples

Turning our attention to Chapter 5, recall that for pairs of rational $q$-commuting unitaries, we have a complete description of the matrix extreme points of the matrix state spaces (both in the rational and irrational cases). In particular, we showed that boundary representations for an operator system generated by $q$-commuting unitaries have

constant dimension. Alas, for $d$-tuples of $\Lambda$-commuting unitaries, we did not have a definitive answer. This raises two natural questions:

1. Given some rational $\Lambda$, what is $N(\Lambda)$?

2. Is there an operator system generated by a rational $\Lambda$-commuting $d$-tuple which admits a boundary representation of dimension lower then $N(\Lambda)$?

Answering these questions will alow us to generalize more results from the $d = 2$ case to larger values of $d$.

# Bibliography


[1] Erik M Alfsen. *Compact Convex Sets and Boundary Integrals*, volume 57. Springer Science & Business Media, 2012.

[2] William B Arveson. Subalgebras of C*-algebras. *Acta Mathematica*, 123:141–224, 1969.

[3] Florin-Petre Boca. *Rotation C*-algebras and almost Mathieu operators*. The Theta Foundation, Bucharest, 2001.

[4] Man-Duen Choi. Positive linear maps on C*-algebras. *Canadian Journal of Mathematics*, 24(3):520–529, 1972.

[5] Man-Duen Choi and Edward G Effros. Injectivity and operator spaces. *Journal of functional analysis*, 24(2):156–209, 1977.

[6] Kenneth R. Davidson. *C*-Algebras by Example*, volume 6. American Mathematical Soc., 1996.

[7] Kenneth R. Davidson, Adam Dor-On, Orr Moshe Shalit, and Baruch Solel. Dilations, inclusions of matrix convex sets, and completely positive maps. *International Mathematics Research Notices*, 2017(13):4069–4130, 2017.

[8] Kenneth R. Davidson and Matthew Kennedy. The choquet boundary of an operator system. *Duke Mathematical Journal*, 164(15):2989–3004, 2015.

[9] Eric Evert, J William Helton, Igor Klep, and Scott McCullough. Extreme points of matrix convex sets, free spectrahedra, and dilation theory. *The Journal of Geometric Analysis*, 28(2):1373–1408, 2018.

[10] Douglas Farenick and Ryan Tessier. Purity of the embeddings of operator systems into their C*-and injective envelopes. *Pacific Journal of Mathematics*, 317(2):317–338, 2022.

[11] Douglas R Farenick. Extremal matrix states on operator systems. *Journal of the London Mathematical Society*, 61(3):885–892, 2000.

[12] Malte Gerhold and Orr Moshe Shalit. Dilations of q-commuting unitaries. *International Mathematics Research Notices*, 2022(1):63–88, 2022.

[13] Malte Gerhold, Satish K Pandey, Orr Moshe Shalit, and Baruch Solel. Dilations of unitary tuples. *Journal of the London Mathematical Society*, 104(5):2053–2081, 2021.

[14] Tom-Lukas Kriel. An introduction to matrix convex sets and free spectrahedra. *Complex Analysis and Operator Theory*, 13(7):3251–3335, 2019.

[15] Richard I Loebl and Vern I Paulsen. Some remarks on C*-convexity. *Linear Algebra and its Applications*, 35:63–78, 1981.

[16] Paul S Muhly and Baruch Solel. An algebraic characterization of boundary representations. In *Nonselfadjoint operator algebras, operator theory, and related topics*, pages 189–196. Springer, 1998.

[17] KR Parthasarathy. *An Introduction to Quantum Stochastic Calculus*, volume 85. Springer Science & Business Media, 1992.

[18] Benjamin Passer and Vern I Paulsen. Matrix range characterizations of operator system properties. *arXiv preprint arXiv:1912.06279*, 2019.

[19] Benjamin Passer, Orr Moshe Shalit, and Baruch Solel. Minimal and maximal matrix convex sets. *Journal of Functional Analysis*, 274(11):3197–3253, 2018.

[20] Vern Paulsen. *Completely Bounded Maps and Operator Algebras*. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 2002.

[21] W Forrest Stinespring. Positive functions on C*-algebras. *Proceedings of the American Mathematical Society*, 6(2):211–216, 1955.

[22] Corran Webster and Soren Winkler. The Krein-Milman theorem in operator convexity. *Transactions of the American Mathematical Society*, 351(1):307–322, 1999.

הרציונלי, מרחבים אלה תמיד נוצרים סופית. במקרה של $d = 2$, אנחנו אף מוצאים בדיוק את $n$ שעבורו המרחב המטריצי הוא $n$-נוצר. בעזרת תוצאות אלו, אנחנו מוכיחים משפט סיווג למערכות אופרטוריות כאלה ומקשרים את התוצאות הללו לבעיות הכלה של קבוצות מטריציות, ולבעיות אינטרפולציה של העתקות חיוביות לחלוטין בין מערכות אופרטוריות מסויימות.

אנחנו משערים כי חקירה מעמיקה יותר של תורת ההצגות של $d$יות $\Lambda$-מתחלפות תאפשר לנו לחדד אף יותר את התוצאות שקיבלנו בחלקו האחרון של החיבור. יתרה מכך, הכלים שפיתחנו מאפשרים להסיק תוצאות רבות על מערכות אופרטוריות בעזרת חקירת מרחב המצבים המטריציים שלהן וזיהוי נקודות הקיצון המטריציות המתאימות. בנוסף, תכונות מסויימות של קבוצות מטריציות קמורות נוצרות סופית, יכולות לשמש בפתרון בעיות הכלה מסויימות וכפועל יוצא מכך, בעיות אינטרפולציה/דילציה שונות.

# תקציר


מערכת אופרטורית $\mathcal{S}$ היא תת-מרחב צמוד לעצמו בתוך אלגברת $C^*$ נתונה. בפרט, ניתן להתבונן ב-$\mathcal{S}$ כמערכת אופרטורית בתוך האלגברה הנוצרת על ידי $C^*(\mathcal{S})$. היות ו-$\mathcal{S}$ יוצרת את האלגברה, אפשר לנסות וללמוד על המבנה האלגברי של $C^*(\mathcal{S})$ על ידי חקירת המבנה הנורמי של המערכת $\mathcal{S}$. כמובן שגם המקרה ההפוך נכון, ואפשר לנסות וללמוד על המבנה של $\mathcal{S}$ מחקירת המבנה האלגברי של $C^*(\mathcal{S})$.

אך בהנתן מערכת אופרטורית כנ"ל, ניתן להתבונן על אובייקט מתמטי נוסף המקושר אליה והוא מרחב המצבים המטריציים $\mathrm{UCP}(\mathcal{S})$. זוהי קבוצה מטריצית קמורה וקומפקטית, ומעבודתם של וובסטר ווינקלר, אנחנו יודעים שניתן לשחזר ממנה את $\mathcal{S}$ על ידי התבוננות ב-$A(\mathrm{UCP}(\mathcal{S}))$ (מרחב הפונקציות המטריציות האפיניות של $\mathrm{UCP}(\mathcal{S})$) וזיהוי טבעי של מרחב זה עם $\mathcal{S}$. מכאן שאפשר לנסות ולחקור את $\mathcal{S}$ מתוך המבנה המטריצי הקמור של $\mathrm{UCP}(\mathcal{S})$, ולהיפך.

למעשה, גם ההיפך הוא הנכון. כלומר, לכל קבוצה מטריצית קמורה וקומפקטית $K$ ניתן לשייך את מרחב הפונקציות המטריציות האפיניות $A(K)$, ולאחר מכן לזהות את $\mathrm{UCP}(A(K))$ בצורה טבעית עם $K$. כלומר, הקשר בין מערכות אופרטוריות למרחב המצבים המטריציים שלהן מאפשר לנו אף לחקור ולהבין טוב יותר קבוצות מטריציות קמורות וקומפקטיות כלליות.

החקירה של מרחבי מצבים מטריציים בעזרת חקירה של המבנה הנורמי של $\mathcal{S}$ אינה חדשה, וכבר בעבודתו של וויליאם ארבסון מ-1969 הונחו רבים מהכלים המשמשים לחקירה מהסוג הזה. הכלים הללו, ביחד עם תיאוריה עשירה לה תרמו רבים בהמשך, ניתן לסווג נקודות קיצון מטריציות של $\mathrm{UCP}(\mathcal{S})$ במונחי הצגות שפה סוף ממדיות של $\mathcal{S}$, ולהשתמש בנקודות אלו כדי "לפרוש" את $\mathrm{UCP}(\mathcal{S})$ במובן מסויים. לאחר הצגת המונחים והשפה בה נשתמש, חלקו הראשון של החיבור יעסוק בביאור רעיונות אלו, ולאחר מכן נשתמש בהם כדי להראות שסיווג מערכות אופרטוריות עד כדי העתקה שומרת סדר לחלוטין שקול לסיווג מרחבי מצבים מטריציים עד כדי הומיאומורפיזם מטריצי אפיני.

בשלב הבא של החיבור, אנו עוברים לדון במשפחה מיוחדת של מערכות אופרטוריות / קבוצות קמורות מטריציות. אלו הן המערכות האופרטוריות אשר מרחב המצבים המטריצי שלהן נוצר סופית. קיימות עבודות (כגון עבודתו של קרייל) שעסקו בסוג כזה של קבוצות קמורות מטריציות, אך אנחנו מרחיבים את חלק מהתוצאות שהתקבלו שם לקונטקסט כללי יותר. בין השאר, אנחנו ניעזר בכך כדי להוכיח שלמערכות אופרטוריות כאלה, סיווגן עד כדי העתקה שומרת סדר לחלוטין שקול לסיווגן עד כדי העתקה ששומרת סדר ברמה $N$ של המבנה המטריצי.

חלקו הבא של החיבור עובר לעסוק במשפחה נוספת של מערכות אופרטוריות, הנוצרות על ידי אוניטרים. בנוסף, אנחנו מתמקדים בהעתקות שומרות סדר המעתיקות את היוצרים האוניטרים של מערכת אחת ליוצרים האוניטרים של מערכת אחרת. אנחנו מראים כי שקילות סדר של היוצרים האוניטרים הללו גוררת שקילות של האלגבראות הנוצרות על ידם. כלומר, המבנה האלגברי של אלגברת ה-$C^*$ הנוצרת על ידי אוניטרים נקבע ביחידות על ידי המבנה הנורמי של המערכות האופרטורית שהם פורשים.

חלקו האחרון של החיבור עוסק בשילוב שני העולמות (קרי, קבוצות מטריציות קמורות נוצרות סופית, ומערכות אופרטוריות הנוצרות על ידי אוניטרים). אנחנו מתבוננים בזוגות של אוניטרים $q$-מתחלפים או $d$יה של אוניטרים $\Lambda$ מתחלפים, ומסווגים תחילה את מרחבי המצבים המטריציים שלהם, ומראים שבמקרה

המחקר בוצע בהנחייתו של פרופסור אור שליט בפקולטה למתמטיקה.

אני מודה לטכניון על התמיכה הכספית הנדיבה בהשתלמותי.

# קבוצות מטריציות קמורות נוצרות סופית וסיווג מערכות אופרטוריות הנוצרות על ידי אוניטרים $\Lambda$-מתחלפים

חיבור על מחקר

לשם מילוי חלקי של הדרישות
לקבלת התואר מגיסטר למדעים במתמטיקה

**רן קירי**

הוגש לסנט
הטכניון — מכון טכנולוגי לישראל
אב התשע"ב חיפה אוגוסט 2022

# קבוצות מטריציות קמורות נוצרות סופית וסיווג מערכות אופרטוריות הנוצרות על ידי אוניטרים $\Lambda$-מתחלפים

רן קירי